Assignment 1: Data Analysis and Machine Learning

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How can we use data analytics and Machine Learning to predict energy usage for this house?

Follow detail instructions in the assignment specification and use these heading for implementation and discussion

In [1]:
```
# chatGPT. Prompt: "reorganize the imports"

# seed the random numbers for reproducibility
# do not remove this line
import warnings
warnings.filterwarnings("ignore", category=FutureWarning)

import random
random.seed(10)

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn import linear_model
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score
from sklearn.model_selection import GridSearchCV

import xgboost as xgb
```

```
import lightgbm as lgb
from catboost import CatBoostRegressor
```

1. Analyse and visualise the data

1.1. Read the dataset

In [2]:
```
orginal_df = pd.read_csv('energydata_complete.csv')
print(orginal_df.shape)
orginal_df.head()
```

```
(14941, 29)
```

Out[2]:

	date	Appliances	lights	T1	RH_1	T2	RH_2	T3	RH_3
0	14/02/2016 0:00	50	10	21.790000	39.900000	20.100000	40.79	21.39	40.59
1	14/02/2016 0:10	50	0	21.790000	39.900000	20.033333	40.73	21.39	40.59
2	14/02/2016 0:20	60	10	21.700000	39.933333	19.890000	40.79	21.39	40.53
3	14/02/2016 0:30	40	0	21.633333	39.860000	19.890000	40.79	21.39	40.59
4	14/02/2016 0:40	60	10	21.600000	39.900000	19.790000	40.79	21.39	40.59

5 rows × 29 columns

In [3]:
```
#Keep the original data isolated
df = orginal_df.copy()
df['date'] = pd.to_datetime(df['date'], dayfirst=True)
df.set_index('date', inplace=True)
df.head()
```

Out[3]:

date	Appliances	lights	T1	RH_1	T2	RH_2	T3	RH_3	T4
2016-02-14 00:00:00	50	10	21.790000	39.900000	20.100000	40.79	21.39	40.59	19.2
2016-02-14 00:10:00	50	0	21.790000	39.900000	20.033333	40.73	21.39	40.59	19.2
2016-02-14 00:20:00	60	10	21.700000	39.933333	19.890000	40.79	21.39	40.53	19.2
2016-02-14 00:30:00	40	0	21.633333	39.860000	19.890000	40.79	21.39	40.59	19.2
2016-02-14 00:40:00	60	10	21.600000	39.900000	19.790000	40.79	21.39	40.59	19.1

5 rows × 28 columns

In [4]:

```
# date time year-month-day hour:minute:second
# Appliances, energy use in Wh
# lights, energy use of light fixtures in the house in Wh
# T1, Temperature in kitchen area, in Celsius
# RH_1, Humidity in kitchen area, in %
# T2, Temperature in living room area, in Celsius
# RH_2, Humidity in living room area, in %
# T3, Temperature in laundry room area
# RH_3, Humidity in laundry room area, in %
# T4, Temperature in office room, in Celsius
# RH_4, Humidity in office room, in %
# T5, Temperature in bathroom, in Celsius
# RH_5, Humidity in bathroom, in %
# T6, Temperature outside the building (north side), in Celsius
# RH_6, Humidity outside the building (north side), in %
# T7, Temperature in ironing room , in Celsius
# RH_7, Humidity in ironing room, in %
# T8, Temperature in teenager room 2, in Celsius
# RH_8, Humidity in teenager room 2, in %
# T9, Temperature in parents room, in Celsius
# RH_9, Humidity in parents room, in %
# To_out, Temperature outside (from Chievres weather station), in Celsius
# Press_mg_hg, Pressure (from Chievres weather station), in mm Hg
# RH_out, Humidity outside (from Chievres weather station), in %
# Wind speed (from Chievres weather station), in m/s
# Visibility (from Chievres weather station), in km
# Tdewpoint (from Chievres weather station), Â°C
# rv1, Random variable 1, nondimensional
# rv2, Random variable 2, nondimensional

# !!!rv1 and rv2 are just for testing the robustness of the model, it shoulc
```

```
# the model should be able to predict the energy use without these variables
print(f"Shape of the df: {df.shape}")
print(f"Columns in the df: {df.columns}")
```

```
Shape of the df: (14941, 28)
Columns in the df: Index(['Appliances', 'lights', 'T1', 'RH_1', 'T2', 'RH_
2', 'T3', 'RH_3', 'T4',
       'RH_4', 'T5', 'RH_5', 'T6', 'RH_6', 'T7', 'RH_7', 'T8', 'RH_8', 'T9',
       'RH_9', 'T_out', 'Press_mm_hg', 'RH_out', 'Windspeed', 'Visibility',
       'Tdewpoint', 'rv1', 'rv2'],
      dtype='object')
```

1.2. Analyse data characteristics

In this section I have done:

- Checking for basic information using pandas functions such as describe(), info()
- Checking for duplication and missing values
- Plot graphs

Here are some general insights:

1. Data size: contains 14941 data points
2. Data type: most of the columns are numeric, except for the "date" as object
3. Data missing: there are no missing data -> high quality dataset
4. Scaling: our Appliances columns can be substituted with log_appliances
5. Relationship: while the appliances might have some relationship with temperatures, humidity, some others climate features and strongly with what time is it(hour); some of the features are not that useful

In [5]:
```
df.describe()
```

Out[5]:

	Appliances	lights	T1	RH_1	T2	R
count	14941.000000	14941.000000	14941.000000	14941.000000	14941.000000	14941.000
mean	96.747206	3.235393	22.030776	39.418672	20.519140	39.935
std	97.297111	7.296554	1.469766	3.880367	2.322643	4.328
min	10.000000	0.000000	18.600000	27.023333	16.200000	20.463
25%	50.000000	0.000000	21.000000	36.790000	18.890000	37.260
50%	60.000000	0.000000	21.823333	38.790000	20.000000	39.900
75%	100.000000	0.000000	22.890000	41.400000	21.830000	42.700
max	900.000000	50.000000	26.260000	57.496667	29.856667	56.026

8 rows × 28 columns

In [6]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 14941 entries, 2016-02-14 00:00:00 to 2016-05-27 18:00:00
Data columns (total 28 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   Appliances   14941 non-null  int64  
 1   lights       14941 non-null  int64  
 2   T1           14941 non-null  float64
 3   RH_1         14941 non-null  float64
 4   T2           14941 non-null  float64
 5   RH_2         14941 non-null  float64
 6   T3           14941 non-null  float64
 7   RH_3         14941 non-null  float64
 8   T4           14941 non-null  float64
 9   RH_4         14941 non-null  float64
 10  T5           14941 non-null  float64
 11  RH_5         14941 non-null  float64
 12  T6           14941 non-null  float64
 13  RH_6         14941 non-null  float64
 14  T7           14941 non-null  float64
 15  RH_7         14941 non-null  float64
 16  T8           14941 non-null  float64
 17  RH_8         14941 non-null  float64
 18  T9           14941 non-null  float64
 19  RH_9         14941 non-null  float64
 20  T_out        14941 non-null  float64
 21  Press_mm_hg  14941 non-null  float64
 22  RH_out       14941 non-null  float64
 23  Windspeed    14941 non-null  float64
 24  Visibility   14941 non-null  float64
 25  Tdewpoint    14941 non-null  float64
 26  rv1          14941 non-null  float64
 27  rv2          14941 non-null  float64
dtypes: float64(26), int64(2)
memory usage: 3.3 MB
```

In [7]:
```
df.isnull().sum()
```

Out[7]:
```
Appliances     0
lights         0
T1             0
RH_1           0
T2             0
RH_2           0
T3             0
RH_3           0
T4             0
RH_4           0
T5             0
RH_5           0
T6             0
RH_6           0
T7             0
RH_7           0
T8             0
RH_8           0
T9             0
RH_9           0
T_out          0
Press_mm_hg    0
RH_out         0
Windspeed      0
Visibility     0
Tdewpoint      0
rv1            0
rv2            0
dtype: int64
```

In [8]:
```
df.duplicated().sum()
```

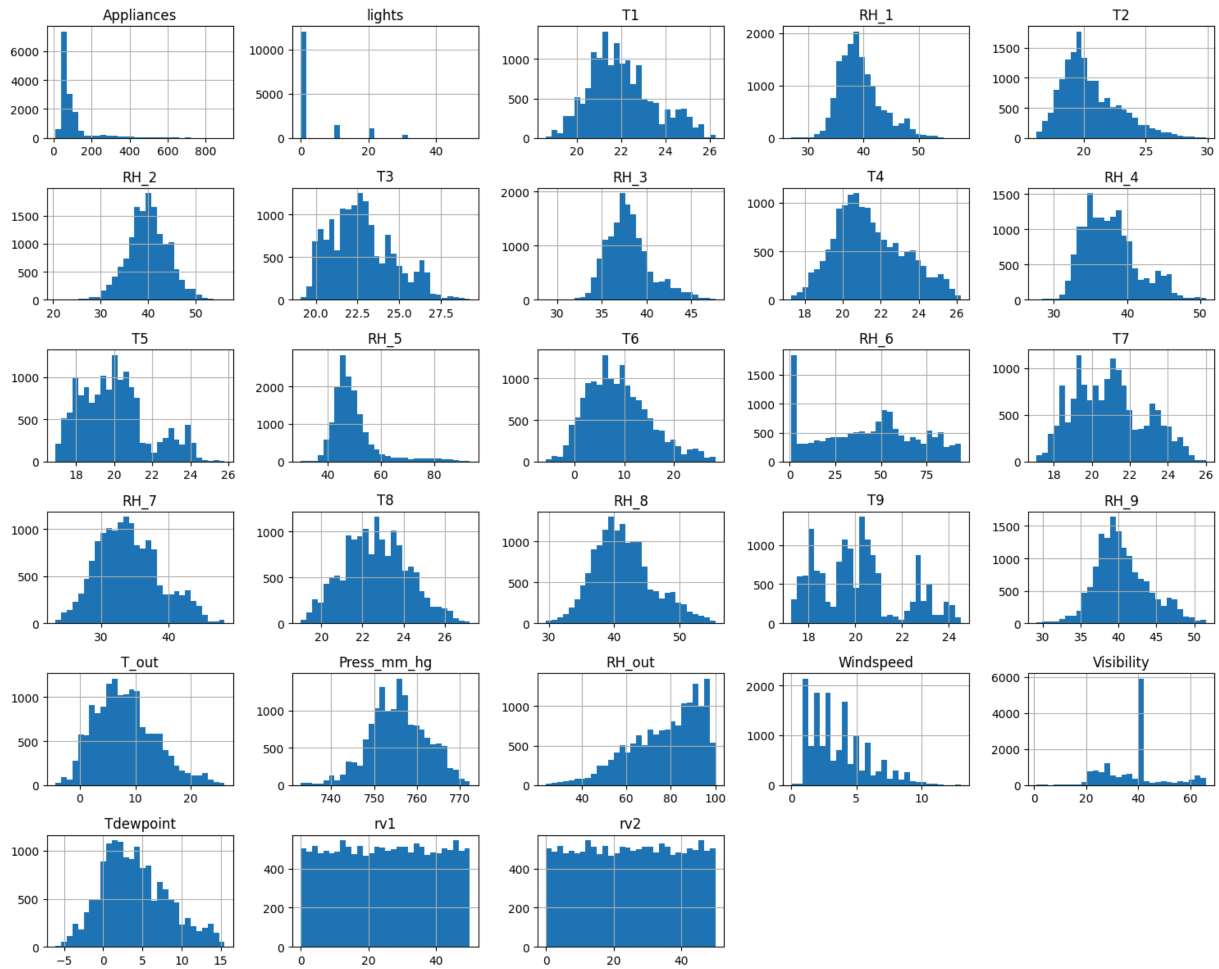
Out[8]:
```
0
```

First, I checked for data completeness. No missing data or duplication -> this dataset is high quality.

In [9]:
```
# chatGPT. Prompt: "plot the histogram"

plt.figure(figsize=(12, 6))
df.hist(figsize=(15, 12), bins=30)
plt.tight_layout()
plt.show()
```

```
<Figure size 1200x600 with 0 Axes>
```



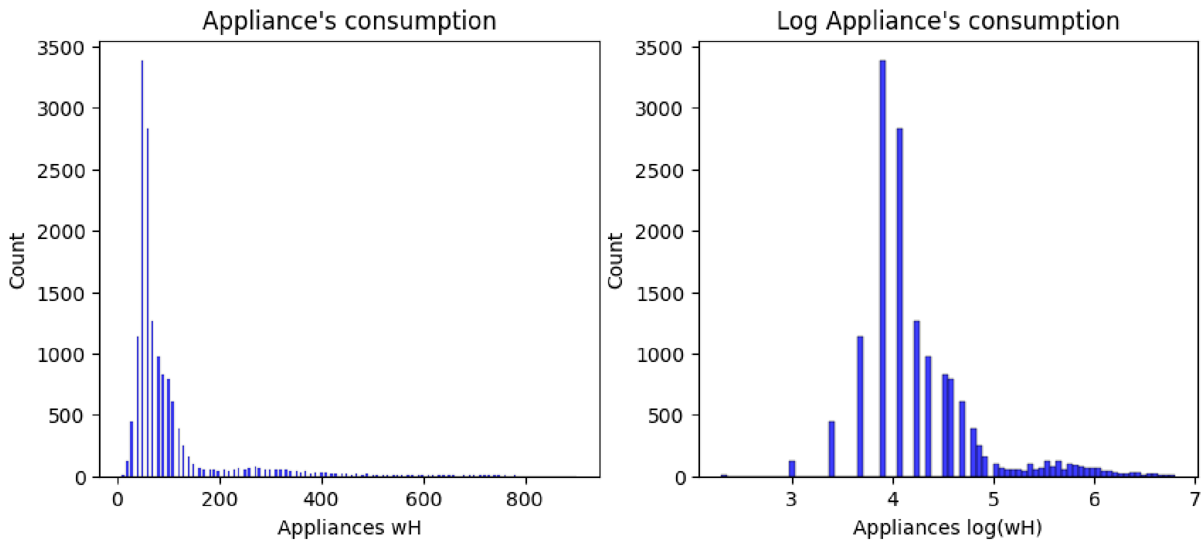
I then chose to plot all of the histogram to have a general idea about each feature's distribution, because by looking at the ***Appliances*** plot, we can see the data is not normally distributed. So that I applied log() to create ***log_appliances*** feature.

In [10]:
```
f, axes = plt.subplots(1, 2, figsize=(10, 4))

sns.histplot(df['Appliances'], element='bars', color='blue', edgecolor='blac
axes[0].set_title("Appliance's consumption")
axes[0].set_xlabel('Appliances wH')

df['log_appliances'] = np.log(df['Appliances'])
sns.histplot(df['log_appliances'], element='bars', color='blue', edgecolor='
axes[1].set_title("Log Appliance's consumption")
axes[1].set_xlabel('Appliances log(wH)')
```

Out[10]: `Text(0.5, 0, 'Appliances log(wH)')`



log_appliances is closer to normal distribution, migitating unstable learning when the target is skewed. From now, we will substitute ***Appliances*** with its log

After multiple iterations of EDA with different time windows, I found ***hour**, *which is extracted from* **date***, is highly correlated with our target.

In [11]:
```
df['hour'] = df.index.hour
df['hour_avg'] = df.groupby('hour')['Appliances'].transform('mean')
df.head()
```

Out[11]:

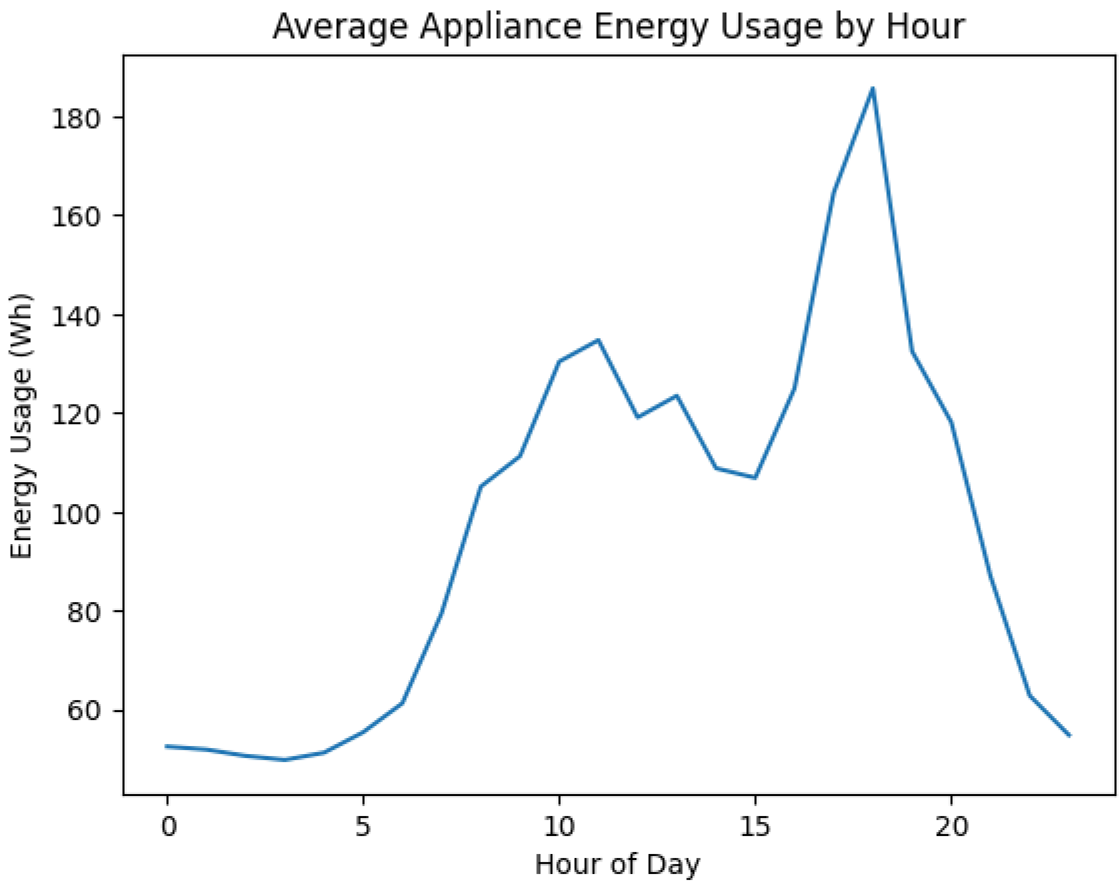
date	Appliances	lights	T1	RH_1	T2	RH_2	T3	RH_3	T4
2016-02-14 00:00:00	50	10	21.790000	39.900000	20.100000	40.79	21.39	40.59	19.2
2016-02-14 00:10:00	50	0	21.790000	39.900000	20.033333	40.73	21.39	40.59	19.2
2016-02-14 00:20:00	60	10	21.700000	39.933333	19.890000	40.79	21.39	40.53	19.2
2016-02-14 00:30:00	40	0	21.633333	39.860000	19.890000	40.79	21.39	40.59	19.2
2016-02-14 00:40:00	60	10	21.600000	39.900000	19.790000	40.79	21.39	40.59	19.1

5 rows × 31 columns

In [12]:
```
daily_energy = df.groupby('hour')['Appliances'].mean()
sns.lineplot(x=daily_energy.index, y=daily_energy.values)
```

```
plt.title('Average Appliance Energy Usage by Hour')
plt.xlabel('Hour of Day')
plt.ylabel('Energy Usage (Wh)')
```

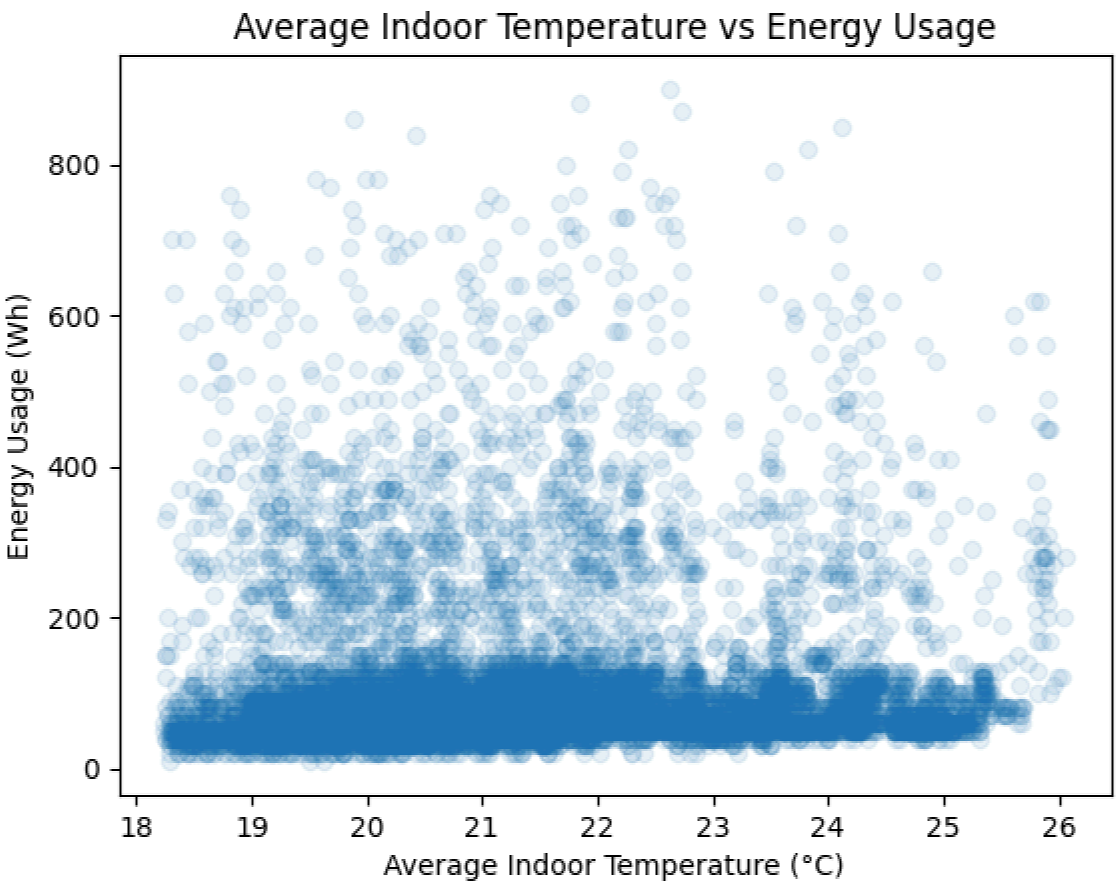
Out[12]: Text(0, 0.5, 'Energy Usage (Wh)')



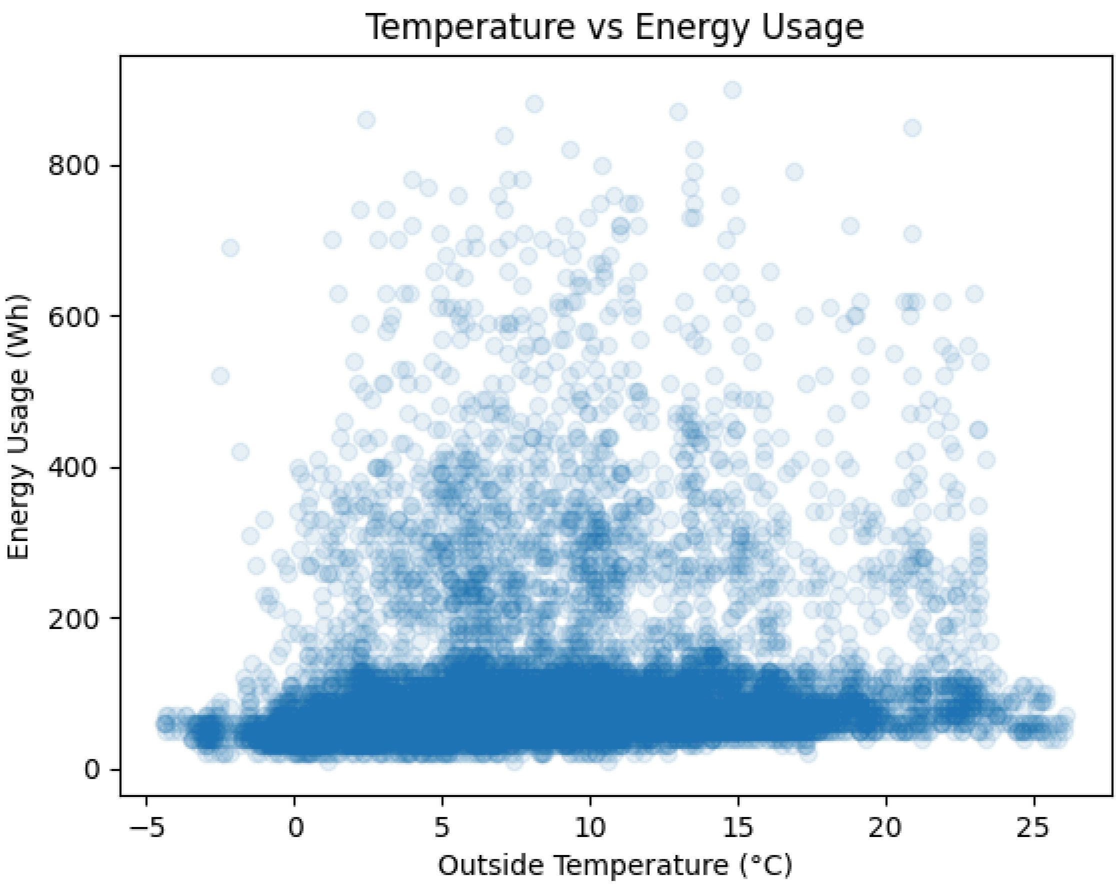
The shown pattern is obvious and understandable. Energy usage peaks at 10 a.m and 6 p.m, and drops for the in-between period of time. This can be attributed to the fact that people might get ready for the day, get out and come back home after work.

In [13]:
```
# avarage inside temp impact on energy usage
avg_indoor_temps = df[['T1', 'T2', 'T3', 'T4', 'T5', 'T7', 'T8', 'T9']].mear
plt.scatter(avg_indoor_temps, df['Appliances'], alpha=0.1)
plt.xlabel('Average Indoor Temperature (°C)')
plt.ylabel('Energy Usage (Wh)')
plt.title('Average Indoor Temperature vs Energy Usage')
```

Out[13]: Text(0.5, 1.0, 'Average Indoor Temperature vs Energy Usage')



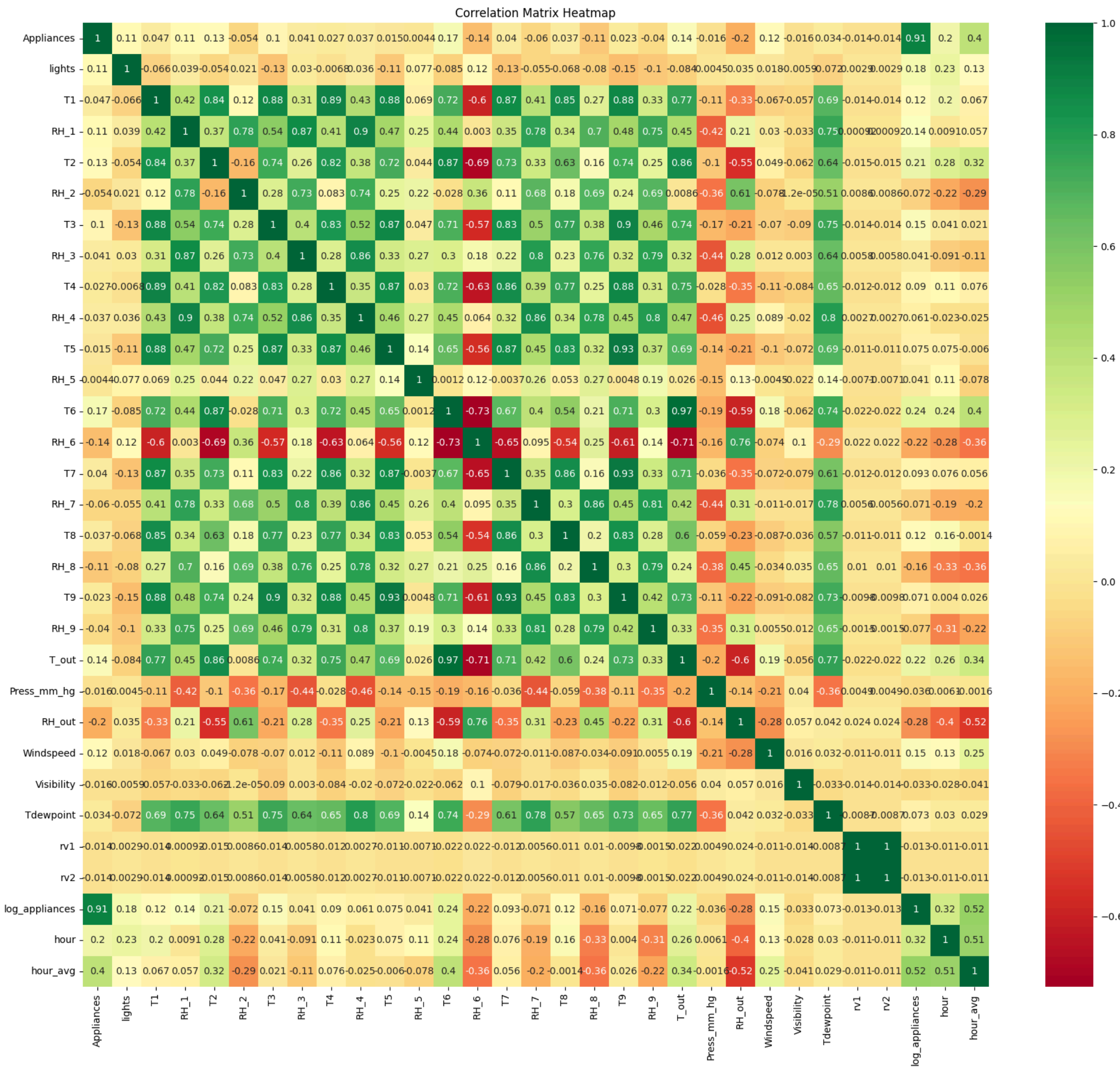
In [14]:
```
# Outside temp impact on energy usage
plt.scatter(df['T_out'], df['Appliances'], alpha=0.1)
plt.xlabel('Outside Temperature (°C)')
plt.ylabel('Energy Usage (Wh)')
plt.title('Temperature vs Energy Usage')
plt.show()
```



At first, I thought temperature might contribute to the amount of energy used because of the need of using more A/C or heater. However, the plots do not show any strong relationship. The wide disperse of those points is showing that there is a non-linearity between those features.

In [15]:
```
# chatGPT. Prompt: "plot the correlation heatmap"

# Correlation Heatmap visualization code
correlation_matrix = df.corr()
plt.figure(figsize=(21, 18))
sns.heatmap(correlation_matrix, annot=True, cmap="RdYlGn")
plt.title("Correlation Matrix Heatmap")
plt.show()
```



Insight from the heatmap:

- All of the temperature indoor and outdoor are correlated -> keep the one that has the highest absolute correlation (***T_6***)
- All of the humidity indoor and outdoor are correlated -> keep the one that has the highest absolute correlation (***RH_6***)
- ***hour*** has the strongest correlation -> ***hour***
- the other climate related features such as ***Press_mm_hg, Windspeed, Visibility, Tdewpoint*** are kept
- ***light*** also has a significant correlation with log_appliances, but the rest is not. So we might conclude that there is no obvious ***linear correlation*** between the target and features, meaning that using Linear Methods might not be good choices.

2. Pre-process the data

In this section I have done:

- combining correlated features for temperature and humidity
- keeping important features based on EDA above
- removing outliers from the dataset
- split the dataset into train, dev, test sets
- scale the dataset

In [16]:
```
# chatGPT. Prompt: "create a new column of average indoor temperature and hu

df['avg_indoor_temp']=df[['T1', 'T2', 'T3', 'T4', 'T5', 'T7', 'T8', 'T9']].m
df['temp_diff']=abs(df['avg_indoor_temp']-df['T_out'])

df['avg_indoor_hum']=df[['RH_1', 'RH_2', 'RH_3', 'RH_4', 'RH_5', 'RH_7', 'RH
#create a column of difference between outside and inside building humidity
df['hum_diff']=abs(df['RH_out']-df['avg_indoor_hum'])

print(df.shape)

df.head()
```

```
(14941, 35)
```

Out[16]:

	Appliances	lights	T1	RH_1	T2	RH_2	T3	RH_3	T4
date									
2016-02-14 00:00:00	50	10	21.790000	39.900000	20.100000	40.79	21.39	40.59	19.2
2016-02-14 00:10:00	50	0	21.790000	39.900000	20.033333	40.73	21.39	40.59	19.2
2016-02-14 00:20:00	60	10	21.700000	39.933333	19.890000	40.79	21.39	40.53	19.2
2016-02-14 00:30:00	40	0	21.633333	39.860000	19.890000	40.79	21.39	40.59	19.2
2016-02-14 00:40:00	60	10	21.600000	39.900000	19.790000	40.79	21.39	40.59	19.1

5 rows × 35 columns

As mentioned above, temperatures and humidities are strongly correlated, I averaged them and only kept the differences between indoor and outdoor in ***temp_diff & hum_diff***

In [17]:
```
numeric_data = df.select_dtypes(include=[np.number])
df_hour = numeric_data.resample('1H').mean()
df_hour.head()
```

Out[17]:

	Appliances	lights	T1	RH_1	T2	RH_2	T3	
date								
2016-02-14 00:00:00	53.333333	6.666667	21.685556	39.898889	19.915556	40.780000	21.390000	4
2016-02-14 01:00:00	53.333333	0.000000	21.410000	39.826667	19.622222	40.897778	21.478611	4
2016-02-14 02:00:00	55.000000	0.000000	21.197222	39.820556	19.392222	41.106667	21.463333	4
2016-02-14 03:00:00	46.666667	0.000000	21.009444	39.938889	19.159444	41.315278	21.397222	4
2016-02-14 04:00:00	58.333333	0.000000	20.812778	40.055000	19.005000	41.518333	21.390000	4

5 rows × 35 columns

In [18]:
```
chosen_columns = ['hour', 'temp_diff', 'hum_diff', 'lights', 'T6', 'RH_6', '
df_hour = df_hour[chosen_columns]
```

In [19]:
```
#Spliting
train_dev_size = int(len(df_hour) * 0.75)

df_train_dev = df_hour.iloc[:train_dev_size]
df_test = df_hour.iloc[train_dev_size:]

print(f"df_train_dev shape: {df_train_dev.shape}")
print(f"df_test shape: {df_test.shape}")
```

```
df_train_dev shape: (1868, 11)
df_test shape: (623, 11)
```

I did not follow the spliting ratio on MyUni because it resulted in a very small training dataset compare to the test set(1500 and 600 rows). I instead devided into roughly 90/10/10. Above is the first splitting, and then remove outliers and then split again later

In [20]:
```
#chatGPT. Prompt: "remove outliers by using IQR method"

# Handling Outliers
for ftr in list(df_train_dev.columns):
  print(f"Removing outliers of {ftr}")
  q_25= np.percentile(df[ftr], 25)
  q_75 = np.percentile(df[ftr], 75)
  iqr = q_75 - q_25
  cut_off = iqr * 1.5
  lower = q_25 - cut_off
  upper = q_75 + cut_off
```

```
  outliers = [x for x in df[ftr] if x < lower or x > upper]

  #removing outliers
  if len(outliers)!=0:

    def bin(row):
      if row[ftr]> upper:
        return upper
      if row[ftr] < lower:
        return lower
      else:
        return row[ftr]

    df_train_dev[ftr] =  df_train_dev.apply (lambda row: bin(row), axis=1)
```

```
Removing outliers of hour
Removing outliers of temp_diff
Removing outliers of hum_diff
Removing outliers of lights
Removing outliers of T6
Removing outliers of RH_6
Removing outliers of Press_mm_hg
Removing outliers of Windspeed
Removing outliers of Visibility
Removing outliers of Tdewpoint
Removing outliers of log_appliances
```

```
/tmp/ipykernel_44194/2747780980.py:25: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/
stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  df_train_dev[ftr] =  df_train_dev.apply (lambda row: bin(row), axis=1)
```

Removing outliers followed these steps:

1. Calculate Q1, Q3
2. Calculate IQR
3. Calculate upper and lower bounds
4. Replace with upper or lower bound

In [21]:
```
splitting_index = int(len(df_train_dev)*0.8)

df_train = df_train_dev.iloc[:splitting_index]
df_dev = df_train_dev.iloc[splitting_index:]

X_train = df_train.drop(columns=['log_appliances'])
y_train = df_train['log_appliances']

X_dev = df_dev.drop(columns=['log_appliances'])
y_dev = df_dev['log_appliances']

X_test = df_test.drop(columns=['log_appliances'])
y_test = df_test['log_appliances']
```

```
print(f"X_train shape: {X_train.shape}")
print(f"y_train shape: {y_train.shape}")
print(f"X_dev shape: {X_dev.shape}")
print(f"y_dev shape: {y_dev.shape}")
print(f"X_test shape: {X_test.shape}")
print(f"y_test shape: {y_test.shape}")
```

```
X_train shape: (1494, 10)
y_train shape: (1494,)
X_dev shape: (374, 10)
y_dev shape: (374,)
X_test shape: (623, 10)
y_test shape: (623,)
```

In [22]:

```
scaler = StandardScaler()
scaler.fit(X_train)
X_train_scaled = scaler.transform(X_train)
X_dev_scaled = scaler.transform(X_dev)
X_test_scaled = scaler.transform(X_test)
```

In [23]:

```
# chatGPT. Prompt: "plot before and after scaling histograms"

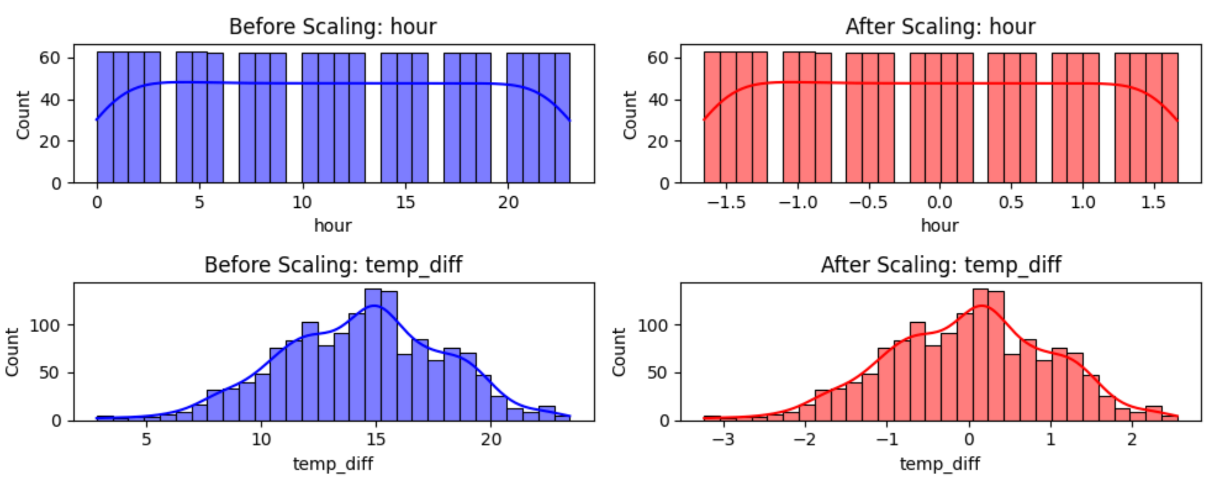
df_not_scaled = pd.DataFrame(X_train, columns=X_train.columns)
df_scaled = pd.DataFrame(X_train_scaled, columns=X_train.columns)

fig, axes = plt.subplots(len(list(df_not_scaled.columns[:2])), 2, figsize=(1

for i, feature in enumerate(list(df_not_scaled.columns[:2])):
    #plotting the not scaled data
    sns.histplot(df_not_scaled[feature], bins=30, kde=True, ax=axes[i, 0], c
    axes[i, 0].set_title(f"Before Scaling: {feature}")

    # Plot the scaled data
    sns.histplot(df_scaled[feature], bins=30, kde=True, ax=axes[i, 1], color
    axes[i, 1].set_title(f"After Scaling: {feature}")

plt.tight_layout()
plt.show()
```



After scaling, features have mean = 0 and std = 1, but still keep the shape

3. Implement, train and select prediction models

In this section, I have tried these models:

1. LinearRegression
2. GLM
3. RandomForestRegressor
4. LGBMRegressor
5. XGBRegressor
6. cbr

RandomForestRegressor yielded the best result!

Linear regression

In [24]:
```
lin_model = linear_model.LinearRegression()
lin_model.fit(X_train_scaled,y_train)
```

Out[24]:
```
▾ LinearRegression
LinearRegression()
```

GLM

In [25]:
```
import statsmodels.api as sm

X_train_glm = sm.add_constant(X_train_scaled)
X_test_glm = sm.add_constant(X_test_scaled)

glm_model = sm.GLM(y_train, X_train_glm, family=sm.families.Gaussian())
glm_results = glm_model.fit()
print(glm_results.summary())
```

```
                 Generalized Linear Model Regression Results
============================================================================
==
Dep. Variable:         log_appliances   No. Observations:                 14
94
Model:                            GLM   Df Residuals:                     14
84
Model Family:                Gaussian   Df Model:
9
Link Function:               Identity   Scale:                         0.217
54
Method:                          IRLS   Log-Likelihood:                -975.
42
Date:                Thu, 27 Feb 2025   Deviance:                       322.
83
Time:                        16:26:58   Pearson chi2:                     32
3.
No. Iterations:                     3   Pseudo R-squ. (CS):             0.26
30
Covariance Type:            nonrobust
============================================================================
==
                 coef    std err          z      P>|z|      [0.025      0.97
5]
----------------------------------------------------------------------------
--
const          4.3012      0.012    356.451      0.000       4.278       4.3
25
x1             0.1795      0.013     13.776      0.000       0.154       0.2
05
x2             0.2900      0.061      4.746      0.000       0.170       0.4
10
x3            -0.1459      0.042     -3.493      0.000      -0.228      -0.0
64
x4         -1.605e-16   2.55e-17     -6.301      0.000    -2.1e-16   -1.11e-
16
x5             0.3426      0.056      6.169      0.000       0.234       0.4
51
x6             0.1064      0.024      4.343      0.000       0.058       0.1
54
x7             0.0032      0.014      0.233      0.816      -0.024       0.0
30
x8             0.0224      0.015      1.528      0.127      -0.006       0.0
51
x9            -0.0069      0.012     -0.562      0.574      -0.031       0.0
17
x10            0.0145      0.046      0.318      0.751      -0.075       0.1
04
============================================================================
==
```

Random Forest

In [26]:
```
rf_model = RandomForestRegressor(n_estimators=100,random_state=1)
rf_model.fit(X_train_scaled, y_train)
```

Out[26]:

```
▾ RandomForestRegressor
RandomForestRegressor(random_state=1)
```

LGBMRegressor

In [27]:

```
model_lgb = lgb.LGBMRegressor(num_leaves=41, n_estimators=200)
model_lgb.fit(X_train_scaled, y_train)
```

```
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of te
sting was 0.000385 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 1777
[LightGBM] [Info] Number of data points in the train set: 1494, number of us
ed features: 9
[LightGBM] [Info] Start training from score 4.301224
```

Out[27]:

```
▾ LGBMRegressor
LGBMRegressor(n_estimators=200, num_leaves=41)
```

XGBRegressor

In [28]:

```
model_xgb = xgb.XGBRegressor(n_estimators=100, learning_rate=0.1, random_sta
model_xgb.fit(X_train_scaled, y_train)
```

Out[28]:

```
▾ XGBRegressor
XGBRegressor(base_score=None, booster=None, callbacks=None,
             colsample_bylevel=None, colsample_bynode=None,
             colsample_bytree=None, device=None, early_stopping_r
ounds=None,
             enable_categorical=False, eval_metric=None, feature_
types=None,
             gamma=None, grow_policy=None, importance_type=None,
             interaction_constraints=None, learning_rate=0.1, max
_bin=None,
```

CatBoostRegressor

In [29]:

```
model_cbr = CatBoostRegressor(random_seed=242, verbose=0, early_stopping_rou
model_cbr.fit(X_train_scaled, y_train)
```

Out[29]:

```
<catboost.core.CatBoostRegressor at 0x7f640bd6b740>
```

In [30]:

```
# Function to evaluate the models
def evaluate(model, test_features, test_labels):
    predictions = model.predict(test_features)
    errors = abs(predictions - test_labels)
```

```
    avg_error = np.mean(errors)
    mape = 100 * np.mean(errors / test_labels)
    r_score = 100*r2_score(test_labels,predictions)
    accuracy = 100 - mape

    return (avg_error, mape, r_score, accuracy)
```

I chose MAPE and R2 score for a number of reasons. MAPE measures the average percentage between predicted and actual values, which is good for comparing across different scales. R2 score explains how well the model align with the variance in the data. I also calculated the accuracy by subtracting 100 by MAPE

In [31]:

```
# chatGPT. Prompt: "evaluate the GLM model"

from sklearn.metrics import mean_absolute_percentage_error, r2_score

y_pred_glm = glm_results.predict(X_test_glm)

avg_error_glm = np.mean(abs(y_pred_glm - y_test))
mape_glm = mean_absolute_percentage_error(y_test, y_pred_glm)*100
r2_glm = r2_score(y_test, y_pred_glm)*100
accuracy_glm = 100 - mape_glm

result_glm = (avg_error_glm, mape_glm, r2_glm, accuracy_glm)
result_glm
```

Out[31]:

```
(0.36928993708497126, 8.582048328445675, 4.902558382602518, 91.417951671554
32)
```

In [33]:

```
result_lin = evaluate(lin_model, X_dev_scaled, y_dev)
result_rf = evaluate(rf_model, X_dev_scaled, y_dev)
result_lgb = evaluate(model_lgb, X_dev_scaled, y_dev)
result_xgb = evaluate(model_xgb, X_dev_scaled, y_dev)
result_cbr = evaluate(model_cbr, X_dev_scaled, y_dev)

# Plotting the results
models = ['Linear Regression', 'GLM', 'Random Forest', 'LightGBM', 'XGBoost'
results = [result_lin, result_glm, result_rf, result_lgb, result_xgb, result

# Extracting metrics
avg_errors = [result[0] for result in results]
mapes = [result[1] for result in results]
r_scores = [result[2] for result in results]
accuracies = [result[3] for result in results]

# Plotting
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# Average Error
sns.barplot(x=models, y=avg_errors, ax=axes[0, 0])
axes[0, 0].set_title('Average Error')
axes[0, 0].set_ylabel('Error')
for i, v in enumerate(avg_errors):
    axes[0, 0].text(i, v + 0.01, f'{v:.2f}', ha='center', va='bottom')
```

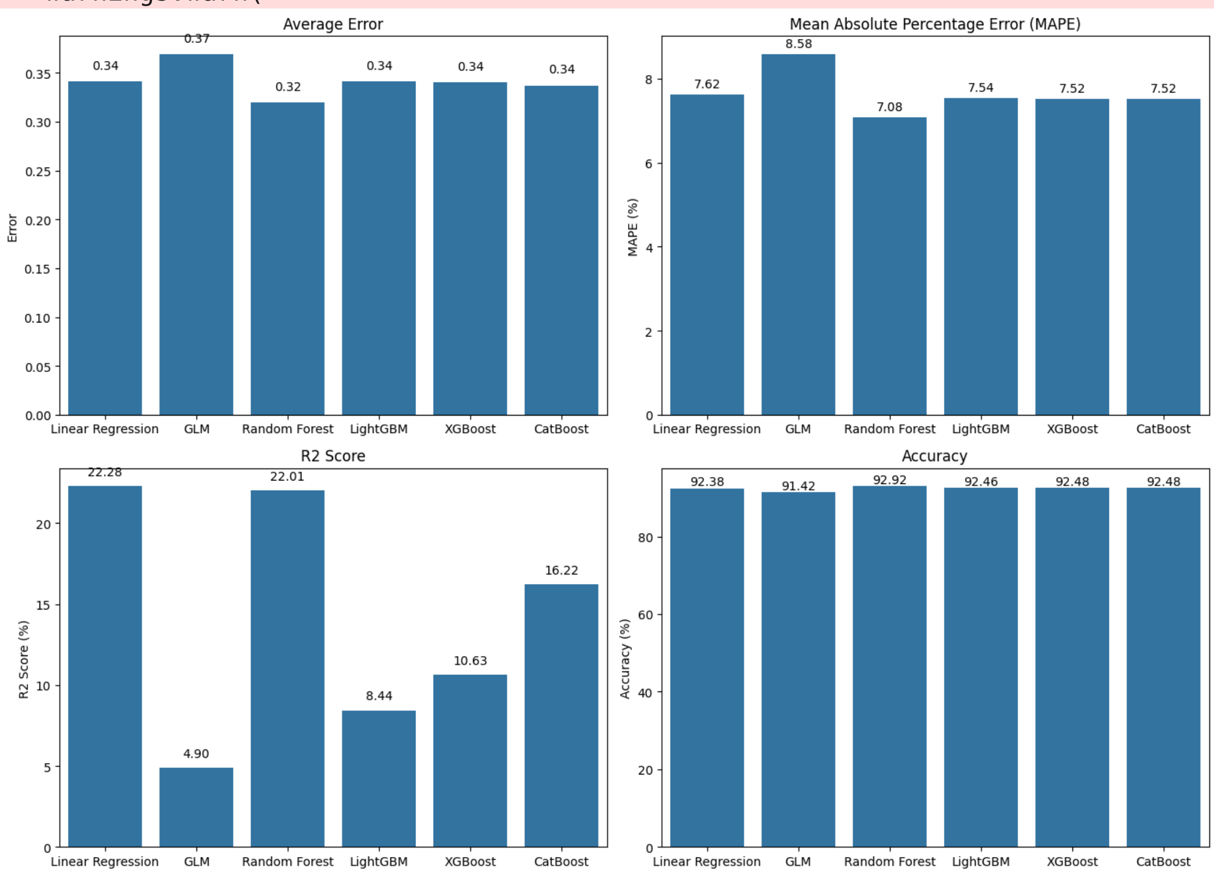
```
# MAPE
sns.barplot(x=models, y=mapes, ax=axes[0, 1])
axes[0, 1].set_title('Mean Absolute Percentage Error (MAPE)')
axes[0, 1].set_ylabel('MAPE (%)')
for i, v in enumerate(mapes):
    axes[0, 1].text(i, v + 0.1, f'{v:.2f}', ha='center', va='bottom')

# R2 Score
sns.barplot(x=models, y=r_scores, ax=axes[1, 0])
axes[1, 0].set_title('R2 Score')
axes[1, 0].set_ylabel('R2 Score (%)')
for i, v in enumerate(r_scores):
    axes[1, 0].text(i, v + 0.5, f'{v:.2f}', ha='center', va='bottom')

# Accuracy
sns.barplot(x=models, y=accuracies, ax=axes[1, 1])
axes[1, 1].set_title('Accuracy')
axes[1, 1].set_ylabel('Accuracy (%)')
for i, v in enumerate(accuracies):
    axes[1, 1].text(i, v + 0.1, f'{v:.2f}', ha='center', va='bottom')

plt.tight_layout()
plt.show()
```

```
/home/james/MyFolder/code/GIT/mbd/.venv/lib/python3.12/site-packages/sklear
n/utils/validation.py:2739: UserWarning: X does not have valid feature name
s, but LGBMRegressor was fitted with feature names
  warnings.warn(
```



As can be seen from the result, especially R2, ***Random Forest Regressor, Linear Regression, XGBoost and CatBoost*** seem to be the best optio. Following is hyper parameter tuning using Grid Search for those models

In [34]:
```
rf_model = RandomForestRegressor(n_estimators=100,random_state=1)
rf_model.fit(X_train_scaled, y_train)

re = evaluate(rf_model, X_test, y_test)
# Define the parameter grid
param_grid = {
    'n_estimators': [100, 200, 300],
    'max_depth': [None, 10, 20, 30],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4],
    'bootstrap': [True, False]
}

# Initialize the GridSearchCV object
grid_search_rf = GridSearchCV(estimator=rf_model, param_grid=param_grid, cv=

# Fit the grid search to the data
grid_search_rf.fit(X_train_scaled, y_train)

# Print the best parameters and the best score
print(f"Best parameters found: {grid_search_rf.best_params_}")
print(f"Best score: {grid_search_rf.best_score_}")
```

```
Fitting 3 folds for each of 216 candidates, totalling 648 fits
```

```
/home/james/MyFolder/code/GIT/mbd/.venv/lib/python3.12/site-packages/sklear
n/utils/validation.py:2732: UserWarning: X has feature names, but RandomFore
stRegressor was fitted without feature names
  warnings.warn(
```

```
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=2, n_estimators=100; total time=   1.3s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=2, n_estimators=100; total time=   1.3s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=2, n_estimators=100; total time=   1.3s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=5, n_estimators=100; total time=   1.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=2, n_estimators=200; total time=   3.0s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=5, n_estimators=100; total time=   1.4s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=2, n_estimators=200; total time=   3.2s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=5, n_estimators=100; total time=   1.2s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=2, n_estimators=200; total time=   3.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=2, n_estimators=300; total time=   4.3s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=5, n_estimators=200; total time=   2.3s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=2, n_estimators=300; total time=   4.6s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=2, n_estimators=300; total time=   4.7s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=5, n_estimators=200; total time=   2.3s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=5, n_estimators=200; total time=   2.9s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=10, n_estimators=100; total time=   1.2s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=10, n_estimators=100; total time=   1.2s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=10, n_estimators=100; total time=   1.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=10, n_estimators=200; total time=   2.0s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=5, n_estimators=300; total time=   4.0s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=5, n_estimators=300; total time=   3.9s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=5, n_estimators=300; total time=   4.8s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=10, n_estimators=200; total time=   2.2s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=2, n_estimators=100; total time=   1.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=2, n_estimators=100; total time=   1.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=10, n_estimators=200; total time=   3.5s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=2, n_estimators=100; total time=   1.3s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=10, n_estimators=300; total time=   3.7s
```

```
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=2, n_estimators=200; total time=   2.3s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=10, n_estimators=300; total time=   4.2s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=2, n_estimators=200; total time=   2.3s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=1, min_samples_spl
it=10, n_estimators=300; total time=   3.7s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=2, n_estimators=200; total time=   2.4s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=5, n_estimators=100; total time=   1.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=5, n_estimators=100; total time=   1.3s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=5, n_estimators=100; total time=   1.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=2, n_estimators=300; total time=   3.6s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=2, n_estimators=300; total time=   3.8s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=5, n_estimators=200; total time=   2.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=5, n_estimators=200; total time=   2.2s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=10, n_estimators=100; total time=   0.9s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=2, n_estimators=300; total time=   4.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=5, n_estimators=200; total time=   2.6s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=10, n_estimators=100; total time=   1.0s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=10, n_estimators=100; total time=   1.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=5, n_estimators=300; total time=   3.7s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=5, n_estimators=300; total time=   4.0s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=10, n_estimators=200; total time=   2.1s[CV] END bootstrap=True, max_dept
h=None, min_samples_leaf=2, min_samples_split=10, n_estimators=200; total ti
me=   2.2s

[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=5, n_estimators=300; total time=   3.6s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=10, n_estimators=200; total time=   2.2s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=2, n_estimators=100; total time=   0.9s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=2, n_estimators=100; total time=   1.2s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=10, n_estimators=300; total time=   3.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=2, n_estimators=100; total time=   1.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
```

```
it=2, n_estimators=200; total time=   2.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=10, n_estimators=300; total time=   3.8s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=2, n_estimators=200; total time=   2.3s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=2, min_samples_spl
it=10, n_estimators=300; total time=   3.2s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=2, n_estimators=200; total time=   2.5s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=5, n_estimators=100; total time=   1.3s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=5, n_estimators=100; total time=   1.2s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=5, n_estimators=100; total time=   1.7s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=5, n_estimators=200; total time=   1.9s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=2, n_estimators=300; total time=   3.2s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=2, n_estimators=300; total time=   3.5s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=2, n_estimators=300; total time=   3.9s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=5, n_estimators=200; total time=   2.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=10, n_estimators=100; total time=   1.4s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=10, n_estimators=100; total time=   1.2s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=5, n_estimators=200; total time=   2.6s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=10, n_estimators=100; total time=   1.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=5, n_estimators=300; total time=   3.1s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=5, n_estimators=300; total time=   3.0s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=5, n_estimators=300; total time=   3.7s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=10, n_estimators=200; total time=   2.3s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=10, n_estimators=200; total time=   2.5s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=10, n_estimators=200; total time=   2.7s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=2, n_estimators=100; total time=   1.3s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=2, n_estimators=100; total time=   1.6s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=10, n_estimators=300; total time=   3.2s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
it=10, n_estimators=300; total time=   3.2s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=2, n_estimators=100; total time=   1.4s
[CV] END bootstrap=True, max_depth=None, min_samples_leaf=4, min_samples_spl
```

```
it=10, n_estimators=300; total time=   3.0s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=5, n_estimators=100; total time=   1.0s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=5, n_estimators=100; total time=   1.5s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=2, n_estimators=200; total time=   2.6s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=2, n_estimators=200; total time=   2.7s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=5, n_estimators=100; total time=   1.0s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=2, n_estimators=200; total time=   3.7s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=2, n_estimators=300; total time=   3.3s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=2, n_estimators=300; total time=   3.7s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=5, n_estimators=200; total time=   2.1s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=2, n_estimators=300; total time=   4.3s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=10, n_estimators=100; total time=   1.0s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=10, n_estimators=100; total time=   1.3s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=5, n_estimators=200; total time=   2.8s
[CV] END bootstrap=True, max_depth=10, min_samples_leaf=1, min_samples_split
=5, n_estimators=200; total time=   3.5s
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[CV] END bootstrap=False, max_depth=30, min_samples_leaf=1, min_samples_spli
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t=5, n_estimators=200; total time=   3.5s
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[CV] END bootstrap=False, max_depth=30, min_samples_leaf=1, min_samples_spli
t=5, n_estimators=300; total time=   6.1s
```

```
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t=2, n_estimators=100; total time=   2.3s
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t=2, n_estimators=100; total time=   1.9s
[CV] END bootstrap=False, max_depth=30, min_samples_leaf=2, min_samples_spli
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[CV] END bootstrap=False, max_depth=30, min_samples_leaf=2, min_samples_spli
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t=5, n_estimators=300; total time=   5.5s
[CV] END bootstrap=False, max_depth=30, min_samples_leaf=2, min_samples_spli
t=10, n_estimators=200; total time=   2.8s
```

```
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```

```
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[CV] END bootstrap=False, max_depth=30, min_samples_leaf=4, min_samples_spli
t=10, n_estimators=300; total time=   3.6s
[CV] END bootstrap=False, max_depth=30, min_samples_leaf=4, min_samples_spli
t=10, n_estimators=300; total time=   3.6s
Best parameters found: {'bootstrap': True, 'max_depth': 10, 'min_samples_lea
f': 4, 'min_samples_split': 10, 'n_estimators': 200}
Best score: -0.19140726112632037
```

In [35]:

```
rf_model = RandomForestRegressor(n_estimators=200,random_state=1, max_depth=
rf_model.fit(X_train_scaled, y_train)

result_rf_train = evaluate(rf_model, X_train_scaled, y_train)
result_rf_dev = evaluate(rf_model, X_dev_scaled, y_dev)

print("Random Forest Model")
print(f"Dev Set: MAPE: {result_rf_dev[1]:.2f}%, R2 Score: {result_rf_dev[2]:
print(f"Train Set: MAPE: {result_rf_train[1]:.2f}%, R2 Score: {result_rf_tra
```

```
Random Forest Model
Dev Set: MAPE: 6.98%, R2 Score: 24.36%, Accuracy: 93.02%
Train Set: MAPE: 4.15%, R2 Score: 75.90%, Accuracy: 95.85%
```

In [36]:

```
# Define the parameter grid for XGBoost
model_xgb = xgb.XGBRegressor(n_estimators=100, learning_rate=0.1, random_sta
model_xgb.fit(X_train_scaled, y_train)

param_grid_xgb = {
    'n_estimators': [100, 200, 300],
    'learning_rate': [0.01, 0.05, 0.1],
    'max_depth': [3, 5, 7],
    'min_child_weight': [1, 3, 5],
}

# Initialize the GridSearchCV object for XGBoost
grid_search_xgb = GridSearchCV(estimator=model_xgb, param_grid=param_grid_xg

# Fit the grid search to the data
grid_search_xgb.fit(X_train_scaled, y_train)

# Print the best parameters and the best score
print(f"Best parameters found: {grid_search_xgb.best_params_}")
print(f"Best score: {grid_search_xgb.best_score_}")
```

```
Fitting 3 folds for each of 81 candidates, totalling 243 fits
[CV] END learning_rate=0.01, max_depth=3, min_child_weight=1, n_estimators=1
00; total time=   0.4s
[CV] END learning_rate=0.01, max_depth=3, min_child_weight=1, n_estimators=1
00; total time=   0.4s
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00; total time=   0.4s
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00; total time=   0.5s
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00; total time=   0.1s
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00; total time=   0.6s
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00; total time=   0.2s
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00; total time=   0.2s
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00; total time=   0.3s
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```

```
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00; total time=   0.3s
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[CV] END learning_rate=0.01, max_depth=5, min_child_weight=1, n_estimators=2
00; total time=   0.7s
[CV] END learning_rate=0.01, max_depth=5, min_child_weight=1, n_estimators=2
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```

```
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00; total time=   0.1s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=1, n_estimators=2
00; total time=   0.3s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=1, n_estimators=2
00; total time=   0.4s[CV] END learning_rate=0.01, max_depth=7, min_child_we
ight=5, n_estimators=200; total time=   1.9s

[CV] END learning_rate=0.05, max_depth=3, min_child_weight=1, n_estimators=2
00; total time=   0.2s
[CV] END learning_rate=0.01, max_depth=7, min_child_weight=5, n_estimators=2
00; total time=   1.7s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=3, n_estimators=1
```

```
00; total time=   0.1s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=1, n_estimators=3
00; total time=   0.3s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=1, n_estimators=3
00; total time=   0.4s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=3, n_estimators=1
00; total time=   0.1s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=1, n_estimators=3
00; total time=   0.5s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=3, n_estimators=1
00; total time=   0.1s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=3, n_estimators=2
00; total time=   0.3s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=3, n_estimators=2
00; total time=   0.3s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=3, n_estimators=2
00; total time=   0.3s
[CV] END learning_rate=0.01, max_depth=7, min_child_weight=5, n_estimators=2
00; total time=   2.1s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=5, n_estimators=1
00; total time=   0.1s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=5, n_estimators=1
00; total time=   0.2s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=3, n_estimators=3
00; total time=   0.6s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=3, n_estimators=3
00; total time=   0.4s[CV] END learning_rate=0.05, max_depth=3, min_child_we
ight=5, n_estimators=100; total time=   0.1s

[CV] END learning_rate=0.01, max_depth=7, min_child_weight=5, n_estimators=3
00; total time=   2.3s
[CV] END learning_rate=0.01, max_depth=7, min_child_weight=5, n_estimators=3
00; total time=   2.3s
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00; total time=   0.2s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=3, n_estimators=3
00; total time=   0.5s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=5, n_estimators=2
00; total time=   0.3s
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00; total time=   2.6s
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00; total time=   0.2s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=5, n_estimators=3
00; total time=   0.3s
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00; total time=   0.3s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=5, n_estimators=3
00; total time=   0.3s
[CV] END learning_rate=0.05, max_depth=3, min_child_weight=5, n_estimators=3
00; total time=   0.4s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=1, n_estimators=1
00; total time=   0.4s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=1, n_estimators=1
00; total time=   0.5s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=3, n_estimators=1
```

```
00; total time=   0.3s[CV] END learning_rate=0.05, max_depth=5, min_child_we
ight=1, n_estimators=200; total time=   0.7s

[CV] END learning_rate=0.05, max_depth=5, min_child_weight=1, n_estimators=2
00; total time=   0.6s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=1, n_estimators=2
00; total time=   0.7s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=3, n_estimators=1
00; total time=   0.6s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=3, n_estimators=1
00; total time=   0.3s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=1, n_estimators=3
00; total time=   0.8s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=1, n_estimators=3
00; total time=   0.9s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=3, n_estimators=2
00; total time=   0.5s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=3, n_estimators=2
00; total time=   0.7s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=5, n_estimators=1
00; total time=   0.3s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=3, n_estimators=2
00; total time=   0.7s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=5, n_estimators=1
00; total time=   0.3s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=1, n_estimators=3
00; total time=   1.2s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=5, n_estimators=1
00; total time=   0.4s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=3, n_estimators=3
00; total time=   0.8s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=3, n_estimators=3
00; total time=   0.9s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=3, n_estimators=3
00; total time=   0.9s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=5, n_estimators=2
00; total time=   0.5s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=5, n_estimators=2
00; total time=   0.5s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=5, n_estimators=2
00; total time=   0.6s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=5, n_estimators=3
00; total time=   1.1s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=5, n_estimators=3
00; total time=   0.9s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=1, n_estimators=1
00; total time=   0.9s
[CV] END learning_rate=0.05, max_depth=5, min_child_weight=5, n_estimators=3
00; total time=   1.1s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=1, n_estimators=1
00; total time=   1.1s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=1, n_estimators=1
00; total time=   1.3s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=1, n_estimators=2
00; total time=   1.6s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=3, n_estimators=1
```

```
00; total time=   1.2s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=3, n_estimators=1
00; total time=   1.2s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=1, n_estimators=2
00; total time=   1.8s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=3, n_estimators=1
00; total time=   0.9s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=1, n_estimators=2
00; total time=   2.7s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=1, n_estimators=3
00; total time=   2.4s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=1, n_estimators=3
00; total time=   2.9s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=3, n_estimators=2
00; total time=   1.5s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=3, n_estimators=2
00; total time=   1.6s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=5, n_estimators=1
00; total time=   0.6s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=3, n_estimators=2
00; total time=   2.0s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=1, n_estimators=3
00; total time=   3.5s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=5, n_estimators=1
00; total time=   0.7s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=3, n_estimators=3
00; total time=   2.4s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=5, n_estimators=1
00; total time=   1.1s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=5, n_estimators=2
00; total time=   1.0s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=1, n_estimators=10
0; total time=   0.1s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=1, n_estimators=10
0; total time=   0.1s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=3, n_estimators=3
00; total time=   2.7s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=1, n_estimators=10
0; total time=   0.1s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=5, n_estimators=2
00; total time=   1.3s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=5, n_estimators=2
00; total time=   1.4s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=1, n_estimators=20
0; total time=   0.3s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=1, n_estimators=20
0; total time=   0.2s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=1, n_estimators=20
0; total time=   0.3s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=3, n_estimators=10
0; total time=   0.1s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=1, n_estimators=30
0; total time=   0.3s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=3, n_estimators=3
00; total time=   2.9s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=1, n_estimators=30
```

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0; total time=   0.3s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=1, n_estimators=30
0; total time=   0.3s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=5, n_estimators=3
00; total time=   1.5s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=3, n_estimators=10
0; total time=   0.1s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=3, n_estimators=10
0; total time=   0.1s
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00; total time=   1.4s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=3, n_estimators=20
0; total time=   0.2s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=3, n_estimators=20
0; total time=   0.2s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=3, n_estimators=20
0; total time=   0.3s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=5, n_estimators=10
0; total time=   0.1s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=3, n_estimators=30
0; total time=   0.3s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=5, n_estimators=10
0; total time=   0.1s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=5, n_estimators=10
0; total time=   0.1s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=3, n_estimators=30
0; total time=   0.3s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=3, n_estimators=30
0; total time=   0.4s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=5, n_estimators=20
0; total time=   0.2s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=5, n_estimators=20
0; total time=   0.2s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=5, n_estimators=20
0; total time=   0.2s
[CV] END learning_rate=0.05, max_depth=7, min_child_weight=5, n_estimators=3
00; total time=   1.6s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=5, n_estimators=30
0; total time=   0.3s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=5, n_estimators=30
0; total time=   0.4s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=1, n_estimators=10
0; total time=   0.3s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=1, n_estimators=10
0; total time=   0.3s
[CV] END learning_rate=0.1, max_depth=3, min_child_weight=5, n_estimators=30
0; total time=   0.4s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=1, n_estimators=10
0; total time=   0.4s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=3, n_estimators=10
0; total time=   0.3s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=1, n_estimators=20
0; total time=   0.6s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=3, n_estimators=10
0; total time=   0.4s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=1, n_estimators=20
```

```
0; total time=   0.6s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=1, n_estimators=20
0; total time=   1.0s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=3, n_estimators=10
0; total time=   0.4s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=1, n_estimators=30
0; total time=   0.8s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=1, n_estimators=30
0; total time=   0.9s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=1, n_estimators=30
0; total time=   0.9s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=3, n_estimators=20
0; total time=   0.5s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=3, n_estimators=20
0; total time=   0.6s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=3, n_estimators=20
0; total time=   0.6s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=5, n_estimators=10
0; total time=   0.2s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=5, n_estimators=10
0; total time=   0.3s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=5, n_estimators=10
0; total time=   0.3s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=5, n_estimators=20
0; total time=   0.4s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=3, n_estimators=30
0; total time=   0.7s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=5, n_estimators=20
0; total time=   0.5s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=3, n_estimators=30
0; total time=   0.9s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=5, n_estimators=20
0; total time=   0.5s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=3, n_estimators=30
0; total time=   0.9s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=5, n_estimators=30
0; total time=   0.7s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=5, n_estimators=30
0; total time=   0.8s
[CV] END learning_rate=0.1, max_depth=5, min_child_weight=5, n_estimators=30
0; total time=   0.7s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=1, n_estimators=10
0; total time=   0.8s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=1, n_estimators=10
0; total time=   0.9s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=1, n_estimators=10
0; total time=   1.1s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=3, n_estimators=10
0; total time=   0.6s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=1, n_estimators=20
0; total time=   1.6s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=3, n_estimators=10
0; total time=   0.8s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=1, n_estimators=20
0; total time=   1.7s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=1, n_estimators=20
```

```
0; total time=   2.0s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=3, n_estimators=10
0; total time=   0.7s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=3, n_estimators=20
0; total time=   1.1s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=1, n_estimators=30
0; total time=   2.2s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=3, n_estimators=20
0; total time=   1.5s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=3, n_estimators=20
0; total time=   1.5s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=5, n_estimators=10
0; total time=   0.6s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=1, n_estimators=30
0; total time=   3.6s[CV] END learning_rate=0.1, max_depth=7, min_child_weig
ht=5, n_estimators=100; total time=   0.5s

[CV] END learning_rate=0.1, max_depth=7, min_child_weight=3, n_estimators=30
0; total time=   1.8s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=1, n_estimators=30
0; total time=   3.5s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=3, n_estimators=30
0; total time=   2.3s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=3, n_estimators=30
0; total time=   1.7s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=5, n_estimators=10
0; total time=   1.0s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=5, n_estimators=20
0; total time=   1.0s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=5, n_estimators=20
0; total time=   1.3s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=5, n_estimators=20
0; total time=   1.1s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=5, n_estimators=30
0; total time=   1.2s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=5, n_estimators=30
0; total time=   1.3s
[CV] END learning_rate=0.1, max_depth=7, min_child_weight=5, n_estimators=30
0; total time=   1.0s
Best parameters found: {'learning_rate': 0.01, 'max_depth': 3, 'min_child_we
ight': 3, 'n_estimators': 300}
Best score: -0.1781627145930429
```

In [37]:

```
model_xgb = xgb.XGBRegressor(n_estimators=300, learning_rate=0.01, random_st
model_xgb.fit(X_train_scaled, y_train)

result_xgb_train = evaluate(model_xgb, X_train_scaled, y_train)
result_xgb_dev = evaluate(model_xgb, X_dev_scaled, y_dev)

print("XGBoost Model")
print(f"Dev Set: MAPE: {result_xgb_dev[1]:.2f}%, R2 Score: {result_xgb_dev[2
print(f"Train Set: MAPE: {result_xgb_train[1]:.2f}%, R2 Score: {result_xgb_t
```

```
XGBoost Model
Dev Set: MAPE: 6.95%, R2 Score: 29.12%, Accuracy: 93.05%
Train Set: MAPE: 6.13%, R2 Score: 48.52%, Accuracy: 93.87%
```

In [38]:
```
model_cbr = CatBoostRegressor(random_seed=242, verbose=0, early_stopping_rou
model_cbr.fit(X_train_scaled, y_train)
# Define the parameter grid for CatBoost
param_grid_cbr = {
    'iterations': [100, 200, 300],
    'depth': [4, 6, 8],
    'learning_rate': [0.01, 0.05, 0.1],
    'l2_leaf_reg': [1, 3, 5]
}

# Initialize the GridSearchCV object for CatBoost
grid_search_cbr = GridSearchCV(estimator=model_cbr, param_grid=param_grid_cb

# Fit the grid search to the data
grid_search_cbr.fit(X_train_scaled, y_train)

# Print the best parameters and the best score
print(f"Best parameters found: {grid_search_cbr.best_params_}")
print(f"Best score: {grid_search_cbr.best_score_}")
```

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Fitting 3 folds for each of 81 candidates, totalling 243 fits
[CV] END depth=4, iterations=100, l2_leaf_reg=1, learning_rate=0.01; total t
ime=   0.3s
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ime=   0.4s
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[CV] END depth=4, iterations=100, l2_leaf_reg=3, learning_rate=0.1; total ti
me=   0.8s
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[CV] END depth=4, iterations=200, l2_leaf_reg=1, learning_rate=0.01; total t
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ime=   0.9s
[CV] END depth=6, iterations=100, l2_leaf_reg=5, learning_rate=0.05; total t
ime=   0.8s
[CV] END depth=6, iterations=100, l2_leaf_reg=5, learning_rate=0.1; total ti
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[CV] END depth=6, iterations=100, l2_leaf_reg=5, learning_rate=0.01; total t
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```

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ime=   4.2s
[CV] END depth=8, iterations=100, l2_leaf_reg=3, learning_rate=0.05; total t
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[CV] END depth=8, iterations=100, l2_leaf_reg=3, learning_rate=0.1; total ti
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ime=   7.3s
[CV] END depth=8, iterations=100, l2_leaf_reg=3, learning_rate=0.1; total ti
me=   7.5s
[CV] END depth=8, iterations=100, l2_leaf_reg=5, learning_rate=0.1; total ti
me=   4.9s
[CV] END depth=8, iterations=100, l2_leaf_reg=5, learning_rate=0.1; total ti
me=   4.4s
[CV] END depth=8, iterations=200, l2_leaf_reg=1, learning_rate=0.01; total t
ime=   9.4s
[CV] END depth=8, iterations=200, l2_leaf_reg=1, learning_rate=0.01; total t
ime=   8.9s
[CV] END depth=8, iterations=200, l2_leaf_reg=1, learning_rate=0.01; total t
ime=  10.2s
[CV] END depth=8, iterations=200, l2_leaf_reg=1, learning_rate=0.05; total t
ime=  10.3s
[CV] END depth=8, iterations=200, l2_leaf_reg=1, learning_rate=0.05; total t
ime=  10.8s
[CV] END depth=8, iterations=200, l2_leaf_reg=1, learning_rate=0.1; total ti
me=  10.0s
[CV] END depth=8, iterations=200, l2_leaf_reg=1, learning_rate=0.05; total t
```

```
ime=  12.2s
[CV] END depth=8, iterations=200, l2_leaf_reg=1, learning_rate=0.1; total ti
me=  11.1s
[CV] END depth=8, iterations=200, l2_leaf_reg=1, learning_rate=0.1; total ti
me=   8.1s
[CV] END depth=8, iterations=200, l2_leaf_reg=3, learning_rate=0.01; total t
ime=  10.4s
[CV] END depth=8, iterations=200, l2_leaf_reg=3, learning_rate=0.01; total t
ime=  10.2s
[CV] END depth=8, iterations=200, l2_leaf_reg=3, learning_rate=0.05; total t
ime=   8.6s
[CV] END depth=8, iterations=200, l2_leaf_reg=3, learning_rate=0.1; total ti
me=   8.9s
[CV] END depth=8, iterations=200, l2_leaf_reg=3, learning_rate=0.05; total t
ime=  10.6s
[CV] END depth=8, iterations=200, l2_leaf_reg=3, learning_rate=0.05; total t
ime=  11.8s
[CV] END depth=8, iterations=200, l2_leaf_reg=3, learning_rate=0.1; total ti
me=   7.7s
[CV] END depth=8, iterations=200, l2_leaf_reg=3, learning_rate=0.01; total t
ime=  13.9s
[CV] END depth=8, iterations=200, l2_leaf_reg=3, learning_rate=0.1; total ti
me=   7.4s
[CV] END depth=8, iterations=200, l2_leaf_reg=5, learning_rate=0.01; total t
ime=   8.6s
[CV] END depth=8, iterations=200, l2_leaf_reg=5, learning_rate=0.05; total t
ime=   7.1s
[CV] END depth=8, iterations=200, l2_leaf_reg=5, learning_rate=0.01; total t
ime=   9.0s
[CV] END depth=8, iterations=200, l2_leaf_reg=5, learning_rate=0.05; total t
ime=   9.1s
[CV] END depth=8, iterations=200, l2_leaf_reg=5, learning_rate=0.01; total t
ime=  10.9s
[CV] END depth=8, iterations=200, l2_leaf_reg=5, learning_rate=0.05; total t
ime=  10.7s
[CV] END depth=8, iterations=200, l2_leaf_reg=5, learning_rate=0.1; total ti
me=   7.6s
[CV] END depth=8, iterations=200, l2_leaf_reg=5, learning_rate=0.1; total ti
me=  10.5s
[CV] END depth=8, iterations=200, l2_leaf_reg=5, learning_rate=0.1; total ti
me=  10.8s
[CV] END depth=8, iterations=300, l2_leaf_reg=1, learning_rate=0.01; total t
ime=  12.1s
[CV] END depth=8, iterations=300, l2_leaf_reg=1, learning_rate=0.01; total t
ime=  15.2s
[CV] END depth=8, iterations=300, l2_leaf_reg=1, learning_rate=0.05; total t
ime=  13.0s
[CV] END depth=8, iterations=300, l2_leaf_reg=1, learning_rate=0.05; total t
ime=  13.4s
[CV] END depth=8, iterations=300, l2_leaf_reg=1, learning_rate=0.01; total t
ime=  15.7s
[CV] END depth=8, iterations=300, l2_leaf_reg=1, learning_rate=0.05; total t
ime=  16.1s
[CV] END depth=8, iterations=300, l2_leaf_reg=1, learning_rate=0.1; total ti
me=  15.4s
[CV] END depth=8, iterations=300, l2_leaf_reg=1, learning_rate=0.1; total ti
```

```
me=  14.7s
[CV] END depth=8, iterations=300, l2_leaf_reg=1, learning_rate=0.1; total ti
me=  13.9s
[CV] END depth=8, iterations=300, l2_leaf_reg=3, learning_rate=0.01; total t
ime=  13.0s
[CV] END depth=8, iterations=300, l2_leaf_reg=3, learning_rate=0.01; total t
ime=  15.5s
[CV] END depth=8, iterations=300, l2_leaf_reg=3, learning_rate=0.05; total t
ime=  13.1s
[CV] END depth=8, iterations=300, l2_leaf_reg=3, learning_rate=0.05; total t
ime=  14.4s
[CV] END depth=8, iterations=300, l2_leaf_reg=3, learning_rate=0.05; total t
ime=  16.2s
[CV] END depth=8, iterations=300, l2_leaf_reg=3, learning_rate=0.01; total t
ime=  17.7s
[CV] END depth=8, iterations=300, l2_leaf_reg=3, learning_rate=0.1; total ti
me=  12.4s
[CV] END depth=8, iterations=300, l2_leaf_reg=3, learning_rate=0.1; total ti
me=  15.2s
[CV] END depth=8, iterations=300, l2_leaf_reg=3, learning_rate=0.1; total ti
me=  13.2s
[CV] END depth=8, iterations=300, l2_leaf_reg=5, learning_rate=0.01; total t
ime=  15.3s
[CV] END depth=8, iterations=300, l2_leaf_reg=5, learning_rate=0.01; total t
ime=  14.8s
[CV] END depth=8, iterations=300, l2_leaf_reg=5, learning_rate=0.05; total t
ime=  13.7s
[CV] END depth=8, iterations=300, l2_leaf_reg=5, learning_rate=0.01; total t
ime=  16.5s
[CV] END depth=8, iterations=300, l2_leaf_reg=5, learning_rate=0.05; total t
ime=  14.0s
[CV] END depth=8, iterations=300, l2_leaf_reg=5, learning_rate=0.1; total ti
me=  10.5s
[CV] END depth=8, iterations=300, l2_leaf_reg=5, learning_rate=0.05; total t
ime=  16.6s
[CV] END depth=8, iterations=300, l2_leaf_reg=5, learning_rate=0.1; total ti
me=   9.8s
[CV] END depth=8, iterations=300, l2_leaf_reg=5, learning_rate=0.1; total ti
me=   7.2s
Best parameters found: {'depth': 4, 'iterations': 100, 'l2_leaf_reg': 3, 'le
arning_rate': 0.05}
Best score: -0.17901703316332562
```

In [39]:
```
model_cbr = CatBoostRegressor(depth=4, iterations=100, l2_leaf_reg=3, learni
model_cbr.fit(X_train_scaled, y_train)

result_xgb_train = evaluate(model_cbr, X_train_scaled, y_train)
result_xgb_dev = evaluate(model_cbr, X_dev_scaled, y_dev)

print("CatBoostRegressor Model")
print(f"Dev Set: MAPE: {result_xgb_dev[1]:.2f}%, R2 Score: {result_xgb_dev[2
print(f"Train Set: MAPE: {result_xgb_train[1]:.2f}%, R2 Score: {result_xgb_t
```

```
0:	learn: 0.5213431	total: 764us	remaining: 75.7ms
1:	learn: 0.5135353	total: 2.77ms	remaining: 136ms
2:	learn: 0.5065898	total: 3.61ms	remaining: 117ms
3:	learn: 0.4991283	total: 4.33ms	remaining: 104ms
4:	learn: 0.4922829	total: 5.2ms	remaining: 98.9ms
5:	learn: 0.4852385	total: 7.18ms	remaining: 113ms
6:	learn: 0.4792683	total: 8.07ms	remaining: 107ms
7:	learn: 0.4740553	total: 8.68ms	remaining: 99.8ms
8:	learn: 0.4683986	total: 9.22ms	remaining: 93.3ms
9:	learn: 0.4647445	total: 9.8ms	remaining: 88.2ms
10:	learn: 0.4614330	total: 10.3ms	remaining: 83.3ms
11:	learn: 0.4581110	total: 10.8ms	remaining: 79.1ms
12:	learn: 0.4536404	total: 11.4ms	remaining: 76ms
13:	learn: 0.4502684	total: 11.9ms	remaining: 72.8ms
14:	learn: 0.4467082	total: 12.4ms	remaining: 70.3ms
15:	learn: 0.4435872	total: 13ms	remaining: 68.4ms
16:	learn: 0.4406006	total: 13.6ms	remaining: 66.6ms
17:	learn: 0.4374046	total: 14.4ms	remaining: 65.5ms
18:	learn: 0.4347892	total: 15ms	remaining: 64ms
19:	learn: 0.4329921	total: 15.5ms	remaining: 62.2ms
20:	learn: 0.4313003	total: 16.3ms	remaining: 61.3ms
21:	learn: 0.4295299	total: 17.3ms	remaining: 61.3ms
22:	learn: 0.4277427	total: 18.1ms	remaining: 60.7ms
23:	learn: 0.4258413	total: 18.7ms	remaining: 59.3ms
24:	learn: 0.4245260	total: 19.3ms	remaining: 57.8ms
25:	learn: 0.4226821	total: 19.8ms	remaining: 56.3ms
26:	learn: 0.4213713	total: 20.3ms	remaining: 54.9ms
27:	learn: 0.4194134	total: 20.8ms	remaining: 53.5ms
28:	learn: 0.4179280	total: 21.3ms	remaining: 52.1ms
29:	learn: 0.4169902	total: 22ms	remaining: 51.4ms
30:	learn: 0.4160476	total: 22.5ms	remaining: 50.2ms
31:	learn: 0.4146569	total: 23.2ms	remaining: 49.4ms
32:	learn: 0.4136766	total: 23.8ms	remaining: 48.3ms
33:	learn: 0.4131475	total: 24.3ms	remaining: 47.2ms
34:	learn: 0.4119296	total: 24.8ms	remaining: 46.1ms
35:	learn: 0.4107841	total: 25.3ms	remaining: 45ms
36:	learn: 0.4096911	total: 25.9ms	remaining: 44.1ms
37:	learn: 0.4086358	total: 26.4ms	remaining: 43.1ms
38:	learn: 0.4077725	total: 26.9ms	remaining: 42.1ms
39:	learn: 0.4071658	total: 27.4ms	remaining: 41.1ms
40:	learn: 0.4065992	total: 27.9ms	remaining: 40.2ms
41:	learn: 0.4057807	total: 28.5ms	remaining: 39.3ms
42:	learn: 0.4045675	total: 29ms	remaining: 38.4ms
43:	learn: 0.4040053	total: 29.5ms	remaining: 37.5ms
44:	learn: 0.4032767	total: 30.1ms	remaining: 36.8ms
45:	learn: 0.4024457	total: 33.2ms	remaining: 39ms
46:	learn: 0.4020543	total: 34.3ms	remaining: 38.7ms
47:	learn: 0.4015924	total: 35.1ms	remaining: 38ms
48:	learn: 0.4011909	total: 35.7ms	remaining: 37.2ms
49:	learn: 0.4004591	total: 36.3ms	remaining: 36.3ms
50:	learn: 0.3995288	total: 36.8ms	remaining: 35.4ms
51:	learn: 0.3991022	total: 37.4ms	remaining: 34.5ms
52:	learn: 0.3986954	total: 38.3ms	remaining: 33.9ms
53:	learn: 0.3977521	total: 39ms	remaining: 33.2ms
54:	learn: 0.3969307	total: 39.6ms	remaining: 32.4ms
55:	learn: 0.3965649	total: 40.2ms	remaining: 31.6ms
```

```
56:	learn: 0.3960432	total: 40.9ms	remaining: 30.8ms
57:	learn: 0.3956139	total: 41.6ms	remaining: 30.1ms
58:	learn: 0.3950867	total: 42.4ms	remaining: 29.5ms
59:	learn: 0.3942885	total: 43.3ms	remaining: 28.8ms
60:	learn: 0.3935438	total: 43.9ms	remaining: 28.1ms
61:	learn: 0.3925019	total: 44.5ms	remaining: 27.3ms
62:	learn: 0.3916938	total: 45.4ms	remaining: 26.6ms
63:	learn: 0.3908159	total: 46.8ms	remaining: 26.3ms
64:	learn: 0.3902098	total: 47.7ms	remaining: 25.7ms
65:	learn: 0.3897825	total: 48.4ms	remaining: 24.9ms
66:	learn: 0.3891660	total: 49.2ms	remaining: 24.2ms
67:	learn: 0.3885419	total: 49.9ms	remaining: 23.5ms
68:	learn: 0.3878748	total: 50.6ms	remaining: 22.7ms
69:	learn: 0.3872272	total: 51.5ms	remaining: 22.1ms
70:	learn: 0.3867543	total: 52.1ms	remaining: 21.3ms
71:	learn: 0.3864997	total: 52.7ms	remaining: 20.5ms
72:	learn: 0.3860948	total: 53.2ms	remaining: 19.7ms
73:	learn: 0.3856561	total: 53.7ms	remaining: 18.9ms
74:	learn: 0.3852423	total: 54.2ms	remaining: 18.1ms
75:	learn: 0.3849890	total: 54.8ms	remaining: 17.3ms
76:	learn: 0.3843982	total: 55.3ms	remaining: 16.5ms
77:	learn: 0.3839619	total: 55.9ms	remaining: 15.8ms
78:	learn: 0.3837649	total: 56.4ms	remaining: 15ms
79:	learn: 0.3834083	total: 57ms	remaining: 14.2ms
80:	learn: 0.3829439	total: 57.5ms	remaining: 13.5ms
81:	learn: 0.3825407	total: 58.1ms	remaining: 12.7ms
82:	learn: 0.3822721	total: 58.6ms	remaining: 12ms
83:	learn: 0.3820173	total: 59.2ms	remaining: 11.3ms
84:	learn: 0.3813785	total: 60.7ms	remaining: 10.7ms
85:	learn: 0.3810710	total: 62ms	remaining: 10.1ms
86:	learn: 0.3801601	total: 62.6ms	remaining: 9.36ms
87:	learn: 0.3798515	total: 63.2ms	remaining: 8.62ms
88:	learn: 0.3791285	total: 63.8ms	remaining: 7.88ms
89:	learn: 0.3784002	total: 64.3ms	remaining: 7.15ms
90:	learn: 0.3776463	total: 64.8ms	remaining: 6.41ms
91:	learn: 0.3770618	total: 65.3ms	remaining: 5.68ms
92:	learn: 0.3765366	total: 65.8ms	remaining: 4.95ms
93:	learn: 0.3761032	total: 66.3ms	remaining: 4.23ms
94:	learn: 0.3754897	total: 66.8ms	remaining: 3.51ms
95:	learn: 0.3753016	total: 67.3ms	remaining: 2.8ms
96:	learn: 0.3750583	total: 67.8ms	remaining: 2.1ms
97:	learn: 0.3746137	total: 68.3ms	remaining: 1.39ms
98:	learn: 0.3736734	total: 68.9ms	remaining: 695us
99:	learn: 0.3731635	total: 69.4ms	remaining: 0us
CatBoostRegressor Model
Dev Set: MAPE: 6.81%, R2 Score: 29.67%, Accuracy: 93.19%
Train Set: MAPE: 6.05%, R2 Score: 50.70%, Accuracy: 93.95%
```

As can be seen from after hyper parameter finetuning result, CatBoost gives us the best result where R2 score is 29.67 and MAPE is 6.81 on the Dev Set. And also R2 score on train set and dev set are closer, which means the model is less overfitting.

4. Test the final model and analyse results

In [41]:
```
#Spliting
train_dev_size = int(len(df_hour) * 0.75)

df_train = df_hour.iloc[:train_dev_size]
df_test = df_hour.iloc[train_dev_size:]

X_train = df_train.drop(columns=['log_appliances'])
y_train = df_train['log_appliances']

X_test = df_test.drop(columns=['log_appliances'])
y_test = df_test['log_appliances']

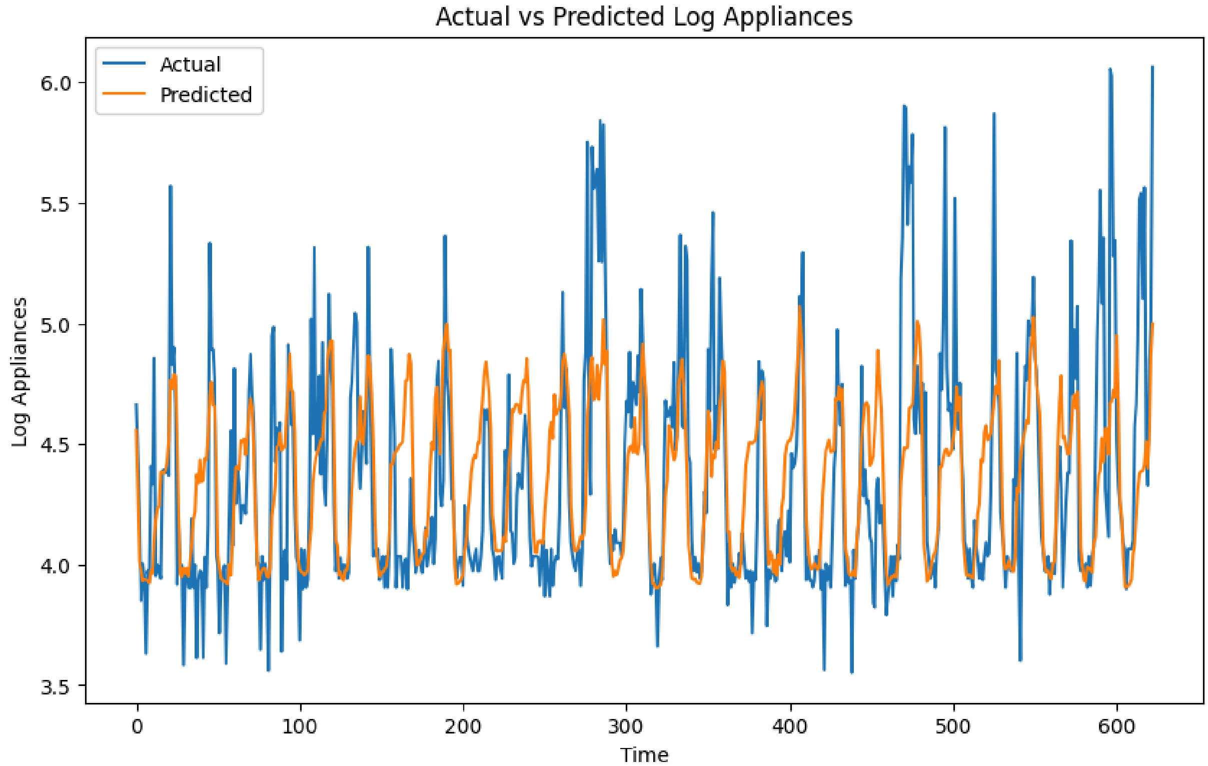
scaler = StandardScaler()
scaler.fit(X_train)
X_train_scaled = scaler.transform(X_train)
X_test_scaled = scaler.transform(X_test)


final_model = CatBoostRegressor(depth=4, iterations=100, l2_leaf_reg=3, lear
final_model.fit(X_train_scaled, y_train)

# Predicting the test set
final_result = evaluate(final_model, X_test_scaled, y_test)
print(f"Final Model: CatBoostRegressor")
print(f"Average Error: {final_result[0]:.2f}")
print(f"MAPE: {final_result[1]:.2f}")
print(f"R2 Score: {final_result[2]:.2f}")
print(f"Accuracy: {final_result[3]:.2f}")
# Plotting the final result
plt.figure(figsize=(10, 6))
plt.plot(y_test.values, label='Actual')
plt.plot(final_model.predict(X_test_scaled), label='Predicted')
plt.xlabel('Time')
plt.ylabel('Log Appliances')
plt.title('Actual vs Predicted Log Appliances')
plt.legend()
plt.show()
```

```
0:	learn: 0.5406201	total: 4.72ms	remaining: 467ms
1:	learn: 0.5334366	total: 6.57ms	remaining: 322ms
2:	learn: 0.5270158	total: 7.18ms	remaining: 232ms
3:	learn: 0.5201510	total: 7.79ms	remaining: 187ms
4:	learn: 0.5136581	total: 8.43ms	remaining: 160ms
5:	learn: 0.5076976	total: 9ms	remaining: 141ms
6:	learn: 0.5015162	total: 9.66ms	remaining: 128ms
7:	learn: 0.4965289	total: 11.3ms	remaining: 130ms
8:	learn: 0.4911892	total: 12.3ms	remaining: 124ms
9:	learn: 0.4869865	total: 13.1ms	remaining: 118ms
10:	learn: 0.4824906	total: 13.8ms	remaining: 111ms
11:	learn: 0.4794107	total: 14.5ms	remaining: 107ms
12:	learn: 0.4756730	total: 15.2ms	remaining: 102ms
13:	learn: 0.4719452	total: 16ms	remaining: 98.2ms
14:	learn: 0.4687836	total: 16.6ms	remaining: 94.3ms
15:	learn: 0.4653783	total: 17.3ms	remaining: 91ms
16:	learn: 0.4621923	total: 18ms	remaining: 88ms
17:	learn: 0.4592539	total: 18.8ms	remaining: 85.7ms
18:	learn: 0.4573204	total: 20ms	remaining: 85.4ms
19:	learn: 0.4553250	total: 21.1ms	remaining: 84.3ms
20:	learn: 0.4540222	total: 21.7ms	remaining: 81.7ms
21:	learn: 0.4520366	total: 22.4ms	remaining: 79.3ms
22:	learn: 0.4502582	total: 23ms	remaining: 77ms
23:	learn: 0.4481736	total: 23.7ms	remaining: 75ms
24:	learn: 0.4464349	total: 24.4ms	remaining: 73.1ms
25:	learn: 0.4447703	total: 25ms	remaining: 71.2ms
26:	learn: 0.4427945	total: 25.8ms	remaining: 69.7ms
27:	learn: 0.4416303	total: 26.5ms	remaining: 68.1ms
28:	learn: 0.4402602	total: 27ms	remaining: 66.2ms
29:	learn: 0.4389846	total: 27.5ms	remaining: 64.3ms
30:	learn: 0.4379517	total: 28.3ms	remaining: 63ms
31:	learn: 0.4368248	total: 29.2ms	remaining: 62ms
32:	learn: 0.4357745	total: 31.5ms	remaining: 63.9ms
33:	learn: 0.4348775	total: 32.5ms	remaining: 63.1ms
34:	learn: 0.4339041	total: 33.4ms	remaining: 62.1ms
35:	learn: 0.4328644	total: 34.2ms	remaining: 60.8ms
36:	learn: 0.4319535	total: 35ms	remaining: 59.5ms
37:	learn: 0.4311309	total: 35.6ms	remaining: 58.2ms
38:	learn: 0.4297482	total: 36.3ms	remaining: 56.8ms
39:	learn: 0.4288180	total: 37ms	remaining: 55.5ms
40:	learn: 0.4276628	total: 37.6ms	remaining: 54.1ms
41:	learn: 0.4271602	total: 38.2ms	remaining: 52.8ms
42:	learn: 0.4264194	total: 38.9ms	remaining: 51.5ms
43:	learn: 0.4257713	total: 39.5ms	remaining: 50.3ms
44:	learn: 0.4247410	total: 40.2ms	remaining: 49.1ms
45:	learn: 0.4236073	total: 40.8ms	remaining: 47.9ms
46:	learn: 0.4230225	total: 41.4ms	remaining: 46.7ms
47:	learn: 0.4221457	total: 42.2ms	remaining: 45.7ms
48:	learn: 0.4213624	total: 43.1ms	remaining: 44.8ms
49:	learn: 0.4203032	total: 43.7ms	remaining: 43.7ms
50:	learn: 0.4197648	total: 44.4ms	remaining: 42.6ms
51:	learn: 0.4188797	total: 45ms	remaining: 41.5ms
52:	learn: 0.4180907	total: 45.7ms	remaining: 40.5ms
53:	learn: 0.4174970	total: 46.3ms	remaining: 39.5ms
54:	learn: 0.4163437	total: 46.9ms	remaining: 38.4ms
55:	learn: 0.4154718	total: 47.5ms	remaining: 37.3ms
```

```
56:	learn: 0.4148231	total: 48.2ms	remaining: 36.3ms
57:	learn: 0.4142727	total: 48.7ms	remaining: 35.3ms
58:	learn: 0.4136761	total: 49.3ms	remaining: 34.3ms
59:	learn: 0.4131001	total: 49.9ms	remaining: 33.3ms
60:	learn: 0.4123950	total: 50.5ms	remaining: 32.3ms
61:	learn: 0.4113681	total: 51.1ms	remaining: 31.3ms
62:	learn: 0.4107510	total: 51.7ms	remaining: 30.4ms
63:	learn: 0.4101977	total: 52.3ms	remaining: 29.4ms
64:	learn: 0.4095274	total: 52.8ms	remaining: 28.5ms
65:	learn: 0.4088569	total: 53.5ms	remaining: 27.5ms
66:	learn: 0.4085528	total: 54.1ms	remaining: 26.6ms
67:	learn: 0.4075063	total: 54.7ms	remaining: 25.8ms
68:	learn: 0.4071572	total: 55.3ms	remaining: 24.8ms
69:	learn: 0.4065340	total: 55.9ms	remaining: 24ms
70:	learn: 0.4061915	total: 56.5ms	remaining: 23.1ms
71:	learn: 0.4055730	total: 57.3ms	remaining: 22.3ms
72:	learn: 0.4051806	total: 58ms	remaining: 21.4ms
73:	learn: 0.4046145	total: 58.9ms	remaining: 20.7ms
74:	learn: 0.4039271	total: 60.7ms	remaining: 20.2ms
75:	learn: 0.4033176	total: 61.7ms	remaining: 19.5ms
76:	learn: 0.4028696	total: 62.6ms	remaining: 18.7ms
77:	learn: 0.4025973	total: 63.4ms	remaining: 17.9ms
78:	learn: 0.4021345	total: 64.1ms	remaining: 17ms
79:	learn: 0.4017404	total: 64.8ms	remaining: 16.2ms
80:	learn: 0.4014651	total: 65.4ms	remaining: 15.3ms
81:	learn: 0.4008634	total: 66.1ms	remaining: 14.5ms
82:	learn: 0.4003861	total: 66.7ms	remaining: 13.7ms
83:	learn: 0.4000303	total: 67.3ms	remaining: 12.8ms
84:	learn: 0.3996139	total: 68.1ms	remaining: 12ms
85:	learn: 0.3988409	total: 68.7ms	remaining: 11.2ms
86:	learn: 0.3984270	total: 69.5ms	remaining: 10.4ms
87:	learn: 0.3980676	total: 70.1ms	remaining: 9.56ms
88:	learn: 0.3978314	total: 71.1ms	remaining: 8.79ms
89:	learn: 0.3973524	total: 71.8ms	remaining: 7.98ms
90:	learn: 0.3968334	total: 72.6ms	remaining: 7.18ms
91:	learn: 0.3964499	total: 73.5ms	remaining: 6.39ms
92:	learn: 0.3961107	total: 74.6ms	remaining: 5.61ms
93:	learn: 0.3957631	total: 76.3ms	remaining: 4.87ms
94:	learn: 0.3955753	total: 77.2ms	remaining: 4.06ms
95:	learn: 0.3954059	total: 77.9ms	remaining: 3.25ms
96:	learn: 0.3945363	total: 79.1ms	remaining: 2.44ms
97:	learn: 0.3939459	total: 79.8ms	remaining: 1.63ms
98:	learn: 0.3933441	total: 80.4ms	remaining: 812us
99:	learn: 0.3927265	total: 81.1ms	remaining: 0us
Final Model: CatBoostRegressor
Average Error: 0.27
MAPE: 5.96
R2 Score: 35.90
Accuracy: 94.04
```



5. Reflection

5.1. Objectives

The objective of this project is developing a machine learning model to predict the energy usage. In order to do that, I have done EDA, Data Processing, and modeling to achieve the best performance. Throughout those main tasks, some mistakes have been made, however, also have been evaluated and resolved. Because of the time limit, there are some ideas about how to further push the results.

5.2. Development process

First, I have done EDA by making some asumption about the dataset. Data visualization techniques tools such as matplotlib and seaborn have been ultilized to extract most valuable insights about the whole dataset or specific features. Then, valuable features will be engineered, or even be introduced to the dataset. Some of the features are less important are aggregated, and some features do not show any correlatiton will be removed. During EDA, I have found that the dataset has a large amount of outliers, which has been handled. Ten features have been kept for scaling because this might help stablizing training process. Only then, the data is ready for training on 6 different models. After evaluating the result, three of the most promising models were picked for further hyper parameters tuning. Finally, I tested the best model on the test set.

5.3. Challenges and Mistakes

During the development process, I have made many challenges and mistakes. EDA is the hardest part in my experiance, because it involves a lot of evaluating technique to be able to get insights out of the plot. At first I did not know where to start, for example, which feature to plot out, which kind of plot to be used, etc. That led to some mistakes related to chosing the features. I dropped ***lights*** column out in the ***version01*** because I chose the wrong plot, I should have looked at the heatmap before deciding which features to be dropped. And I also split the data incorrectly at ***version4*** with different ratio from what on MyUni. One more mistake I have made is fit for scaling before spliting.

5.4. Evaluation and Adaptation

After carefully evaluating the result, I have changed the code and get the desired result without any bug. Data leaking is the most concerning part in my project, where I had to pay attention about when do feature engineering, scaling, and data splitting.

5.5. Potential improvements

There are some potential improments such as using doing more EDA, trying different data preprocessing methods and trying more complex model (Neural network). And if I have more time, I might try to read more academic papers to understand the process more deeply.

6. References

- Candanedo, LM, Feldheim, V & Deramaix, D 2017, ‘Data driven prediction models of energy use of appliances in a low-energy house’, Energy and Buildings, vol. 140, pp. 81–97.
- Bharati, S, Podder, P & Mondal, MRH 2020, ‘Visualization and prediction of energy consumption in smart homes’, International Journal of Hybrid Intelligent Systems, pp. 1–17.
- Brownlee, J 2020, How to Scale Data With Outliers for Machine Learning, Machine Learning Mastery.

7. Appendix

I have used chatGPT in this project. Used prompts have been provided